ONLINE APPOINTMENT SYSTEM FOR AMG DENTAL CLINIC: COMMONWEALTH BRANCH

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Fairview

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

Lloyd Theo T. Abad
Julian Paul S. Alzate
Timothy Ariel M. Basilio
John Francis M. Gayo

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **Online Appointment System for AMG Dental Clinic: Commonwealth Branch**

NAME OF PROPONENTS: Lloyd Theo T. Abad
Julian Paul S. Alzate
Timothy Ariel M. Basilio
John Francis M. Gayo

In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

ENDORSED BY:

Juan Miguel V. Crame
Capstone Project Adviser

APPROVED FOR PROPOSAL DEFENSE:

Christina T. Aseoche
Capstone Project Coordinator

NOTED BY:

Leodigario R. Bongco
Program Head

APRIL 2026

APPROVAL SHEET

This capstone project proposal titled **Online Appointment System for AMG Dental Clinic: Commonwealth Branch**, prepared and submitted by **Lloyd Theo T. Abad, Julian Paul S. Alzate, Timothy Ariel M. Basilio**, and **John Francis M. Gayo**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

Juan Miguel V. Crame
Capstone Project Adviser

Accepted and approved by the Capstone Project Review Panel
in partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

Justin B. Reonisto
Panel Member

Rubinato G. Lubian III
Panel Member

Christina T. Aseoche
Lead Panelist

Noted:

Christina T. Aseoche
Capstone Project Coordinator

Leodigario R. Bongco
Program Head

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INTRODUCTION

Project Context

In the 21st century, dental clinics and other healthcare organizations face growing pressure to replace outdated manual systems with modern digital solutions. Traditionally, clinics relied on pen-and-paper methods to record patient information, schedule appointments, and store physical records in filing cabinets—often leading to misplaced files, retrieval difficulties, and inaccuracies. These manual processes have made it challenging to efficiently track appointments, billing, and treatment history as patient volumes grow. Digital transformation in oral healthcare is becoming essential for improving workflow, reducing administrative errors, and enhancing patient care *(De Ahumada Servant et al., 2024)*. Furthermore, digitalization in dental operations is now considered a strategic necessity for clinics seeking efficiency, accuracy, and competitiveness in the evolving healthcare landscape *(Naamati-Schneider et al., 2024)*.

This is also the case for AMG Dental Clinic: Commonwealth Branch, the client, a private dental clinic founded by Mrs. Michelle A. Macaisa in 2002, which operates from Monday to Saturday, 10:00 AM to 4:00 PM, with two dentists and two assistants *(see Appendix E, p. 80, 81, & 88, No. 1, 3, & 24.2)*. Depending on dentist availability, the client receives approximately 10 to 20 patient inquiries daily across multiple messaging platforms and visits from around 6 to 16 patients per day. The client offers services such as consultations, extractions, orthodontics, and general dental procedures for both regular and HMO patients *(see Appendix E, p. 80, 81, & 88, No. 2, 4, & 25)*. At present, the client manages appointment scheduling and patient record handling through manual procedures handled by the dental assistants *(see Appendix E, p. 80 & 83, No. 3 & 12)*. These front desk responsibilities include responding to patient inquiries, verifying HMO information, and retrieving patient records manually during daily operations *(see Appendix E, p. 81, 84, & 89, No. 4, 15, & 26)*.

One of the current challenges experienced by the client is the scattered mode of communication used in appointment scheduling and patient inquiries. Patients reach the client through different messaging platforms such as SMS, Messenger, and Viber, depending on which platform they prefer *(see Appendix E, p. 81 & 83, No. 5 & 12)*. Scheduling is further delayed as the client relies on asynchronous messaging where patient replies can take several hours and HMO verification may take days or even weeks, and when assistants are occupied during procedures, incoming inquiries are left unattended until one becomes available *(see Appendix E, p. 90, No. 27 & 28)*. These delays have also resulted in lost patients, particularly HMO cases where the extended verification period causes patients to disengage before their appointment is confirmed *(see Appendix E, p. 91, No. 31 & 31.1)*. Communication also becomes fragmented when patients switch platforms mid-transaction, requiring assistants to manually trace prior conversations before the booking can continue *(see Appendix E, p. 91, No. 29)*. Although the client uses a Google calendar with color-coded appointment statuses— blue for unconfirmed, pink for confirmed, and grey for cancelled— scheduling conflicts can still occur since all assistants and dentists are authorized to handle appointment scheduling across these platforms simultaneously *(see Appendix E, p. 89, No. 26 & 26.1)*. With multiple staff managing the same scheduling data across disconnected platforms, duplicate or conflicting bookings become more likely, as demonstrated by instances where one assistant had already booked a patient while another scheduled the same patient at a different time, with similar conflicts arising when cancellations and rescheduling requests are handled by different assistants with no visibility into the appointment's prior status *(see Appendix E, p. 82, 89 & 92, No. 9, 26.2 & 30.1)*. On top of this, appointment reminders are sent manually for every scheduled patient across these same platforms, adding to the overall time spent managing the scheduling process *(see Appendix E, p. 93, No. 33)*.

Another challenge experienced by the client is the manual patient record keeping and retrieval process. Patient information is recorded in paper booklets and stored alphabetically in filing cabinets, and since the client's policy requires all records to be retained indefinitely regardless of the patient's status or the age of the record, the volume of physical files continues to grow over time *(see Appendix E, p. 93 & 96, No. 34 & 40).* As the volume increases, the client has responded by adding more filing cabinets, but this only expands the physical space that must be searched during retrieval, increasing the likelihood of misplacement and making the overall filing arrangement harder to maintain *(see Appendix E, p. 96, No. 40 & 40.1).* Even with alphabetical organization, locating a record can take 5 to 10 minutes, particularly when multiple patients share similar surnames, and with approximately 20,000 records currently on file, misplacement into the wrong alphabetical section is inevitable, requiring a manual search across multiple sections before a patient can proceed with treatment *(see Appendix E, p. 93 - 95, No. 34.1, 36 & 39.1).* The retrieval process is further complicated when follow-up records are temporarily stored outside the cabinets — pulled in advance for scheduled patients — and some of these records are stacked in front of the arranged folders, obstructing access to the files behind them, and when appointments are cancelled, these records must be manually returned and re-alphabetized, and in cases where multiple booklets are handled simultaneously, records are at greater risk of being inserted into the wrong section *(see Appendix E, p. 84, 94 & 95 No. 15, 37, 37.1 & 38).* Physical records are also inherently vulnerable to damage such as water exposure, and since no backup copies exist, any damaged file results in permanent loss of patient information, making it difficult to continue care without manually recollecting the missing details *(see Appendix E, p. 94, No. 35; see Image 7, File Rack).* Collectively, these challenges require dental assistants to manually search through thousands of records, re-alphabetize misfiled booklets after cancellations, and recollect lost patient information from damaged files, all of which consume significant time before a patient can even proceed with treatment.

Another challenge experienced by the client is the manual billing and payment tracking process. After each procedure, the dentist finalizes the cost and presents the patient with a breakdown of fees, after which the assistant collects the payment and records the transaction *(see Appendix E, p. 85, No. 17).* For patients who pay in full, the transaction is logged in the monitoring sheet with only the patient name and total amount paid, while partial payments for installment-based treatments such as orthodontic procedures are recorded exclusively through paper notes attached inside the patient booklet and are not logged in the monitoring sheet until the entire balance is fully settled *(see Appendix E, p. 85 & 101, No. 18, 19, & 56)*. For installment-based treatments, the client requires a 50% down payment upon the start of treatment, with the remaining balance divided across future adjustment visits and updated inside the booklet after every partial payment, and since outstanding balances are not actively tracked between visits, the client has no means to identify or follow up on patients with unpaid balances until they physically return to the clinic *(see Appendix E, p. 99 - 101, No. 48, 49, 50, & 54)*. Recording transactions is further complicated by the variety of payment methods accepted, as errors in noting the payment method used can cause imbalances in the day's financial records, and verifying cashless transactions requires additional time when proof of payment is misplaced or unclear *(see Appendix E, p. 85, No. 19)*. At the end of each day, the client cross-references each transaction by comparing the dentist's computations against the monitoring sheet, and since payments made directly to the head dentist are not immediately reflected in the assistants' records, the monitoring sheet reflects an incorrect total until the head dentist communicates the transaction to the staff, leaving verification incomplete and extending the time needed to close out the day's billing records *(see Appendix E, p. 101, No. 55 & 57)*.

Overall, the client’s reliance on multiple disconnected tools results in inefficiencies, errors, and delays in accessing or updating critical data. These issues affect assistants, dentists, and patients, who may experience longer waiting times and miscommunication. To address these challenges the proposed system aims to centralize operations by streamlining scheduling, managing patient records, simplifying billing, and integrating all functions into a single platform to improve efficiency, reliability, and patient experience.

Purpose and Description

The purpose of this study is to help the client address the inaccuracies and delays in its current operations. Its impact is to make appointment management, patient record keeping, and billing and payment tracking more organized and reliable by bringing all three into one place, reducing the burden on staff that comes from handling these responsibilities through separate and manual means.

The significance of this study is most immediate in how the client manages its appointments. Patients currently reach the client through different messaging platforms, and the independent handling of a shared calendar between two assistants has resulted in conflicting bookings, miscommunication, and prolonged waiting times before appointments can be confirmed. Having all appointment activity managed in one place spares staff from tracing conversations across multiple channels and reduces the risk of conflicts that arise when schedules are updated separately, giving the client a more dependable way of handling bookings from the moment a request is made to the point of confirmation.

It will also ease the difficulties that come with the client's manual patient record keeping, where retrieving physical booklets from filing cabinets can take up to ten minutes per record, misplaced files cause further delays in patient processing, and damaged documents risk the permanent loss of patient information with no means of recovery. Having patient information stored and organized in one place reduces the time spent on retrieval, eliminates the storage limitations that come with a growing volume of physical records, and ensures that patient information remains intact and accessible whenever it is needed.

It will further reduce the recurring inconsistencies in the client's billing and payment tracking, where balances are manually written and updated, transactions made directly to the dentist without the assistant's immediate knowledge leave records temporarily inaccurate, and resolving end-of-day discrepancies requires staff to manually recheck each transaction. Having payment records maintained in one place ensures that balances are always reflected accurately, reducing the time and effort spent on repeated verification and minimizing the inconsistencies that arise when billing is handled through manual means.

It will also serve as a centralized means of keeping the client's appointment activity, patient information, and payment records all in one place, ensuring that every transaction and update is consistently reflected across the client's operations. This reduces the risk of mismatched or outdated records that arise when these responsibilities are managed separately, allowing staff to work from accurate information at any point during the day and to spend less time correcting errors and more time attending to patients.

Objectives

Through interviews with client staff, the proponents identified that the client's current system relies on traditional methods for booking appointments, record-keeping, and billing. In response, the proponents aim to develop an online appointment system to enhance clients' business processes by centralizing scheduling, patient records, and payment tracking into a single platform. To accomplish this, the project will focus on the following objectives:

- **To Develop a Centralized Appointment Scheduling Module**

  This module addresses the inefficiencies caused by their current process, which involves manually checking appointments across multiple platforms. It aims to provide patients with a convenient and accessible way to book appointments at any time without needing to visit or call the clinic. Additionally, it enables assistants and dentists to create, update, approve, reschedule, and monitor appointments while ensuring that all schedule changes are properly reflected in the system. The module organizes appointment details, including patient information, assigned dentist, date, time, and status (pending, confirmed, rescheduled, cancelled, or completed). It helps reduce scheduling conflicts, duplicate bookings, and miscommunication caused by manual or multi-platform appointment handling, while improving the accuracy and efficiency of clinic scheduling operations.

- **To Develop a Digitized Patient Record Module that will secure storage, retrieval, and updating of patient information.**

  This module addresses the challenges of paper-based record-keeping while also reducing the risk of data loss due to fragmented storage. It aims to digitize and organize patient-related information, including personal details and medical/dental histories, into a centralized database. Additionally, it enables authorized users, such as administrators, dentists, and assistants, to securely store, view, and update patient profiles, medical histories, treatment records, and appointment histories. The system features search and filtering functions for faster retrieval of patient records, reducing delays caused by manual filing and misplacement of documents. It also implements role-based access control to ensure that only authorized personnel can access sensitive patient information, improving data security, accuracy, and confidentiality within the system.

- **To Design and Implement a Billing and Invoice Module that will automate the recording of patient payments and monitor outstanding balances.**

  This module addresses the limitations of manual billing, which are prone to errors and inconsistencies, as well as the difficulties in tracking outstanding patient balances. It aims to streamline the clinic's billing processes by allowing authorized users to encode services rendered, calculate corresponding fees, and generate digital invoices for each transaction. The module records essential payment details—such as services availed, amount paid, payment method, and transaction date—to ensure accurate and consistent documentation. It also supports installment-based payment tracking by automatically updating patient balances after each transaction and reflecting the current payment status. Furthermore, the module provides access to financial records, including invoice history and outstanding balances, to support monitoring of clinic revenue. Reducing manual computation errors minimizes discrepancies in recorded payments, which improves the accuracy and reliability of financial record-keeping.

Scope and Limitations

Scope

The system covers the core administrative operations of the client, serving four types of users — the clinic owner, dentists, assistants, and patients. It encompasses appointment scheduling, patient record management, and billing and payment tracking, all handled within a single platform. The system is intended for use within the client's daily operations, from the time a patient books an appointment through the completion of their treatment and payment.

Access Types:

To operate the Online Appointment System properly, there are different types of users to manage this, and each user has specific modules and responsibilities to support the client operations. The system's user types are listed as follows:

- **Admin**

  The Admin account belongs to the owner, who serves as the highest-level user in the proposed study, responsible for overseeing overall clinic operations, including monitoring daily appointment activity, reviewing clinic performance, and managing staff accounts. The owner has no plans to delegate administrative responsibilities to any other staff member *(see Appendix E, p. 90, No. 26.3).*

- **Dentist**

  The Dentist account is responsible for managing their assigned patients and documenting treatment information. This role focuses on the clinical side of operations, giving dentists access to appointment schedules, patient records, and tools for recording diagnoses and treatment plans. Dentists may also manage availability depending on the employment arrangement with the client.

- **Assistant**

  The Assistant account is responsible for managing the day-to-day administrative workflows of the client. This role oversees appointment requests, maintains patient records, and handles billing and payment tracking as patients complete treatments. The assistant also serves as the primary point of contact for appointment-related updates, ensuring that patients are informed of any changes to scheduled appointments.

- **Patient**

  The Patient account is used for booking appointments through the system. A patient can register for an account, log in, select available appointment slots, and proceed with booking based on the dentist assigned to that schedule. A preferred dentist may be selected when both are available, but when only one dentist is available, the booking follows the available schedule. Appointment status can be tracked, and personal information may be updated. Notifications regarding confirmations, reschedules, or reminders are received through email.

Admin’s side

- **Admin Dashboard**

  The Dashboard Module provides the admin with an overview of daily operations and analysis in the system. This module serves as the admin’s main control panel for scheduled appointments and evaluating clinic activity at a glance.

  - **Appointment Schedule**

    This feature displays a table of all requested and confirmed appointments. It is view-only and includes relevant appointment details to help the admin monitor scheduling activity across the clinic.

    - **Filter**

      This feature allows the admin to filter the appointment schedule by day, week, or month for a more focused view of scheduling activity.

  - **Statistics**

    This feature presents visual summaries of the clinic’s performance. It includes a revenue line graph that can be viewed daily, monthly, or yearly as well as data showing the number of new patients and total appointments compared to the previous month.

    - **Revenue Graph View**

      This feature displays a monthly line graph of the clinic's revenue that the admin can examine to track earnings and monitor financial performance over time.

    - **Performance Comparison Card**

This feature displays the number of new patients and total appointments, allowing the admin to compare current figures against the previous day, week, month, or year depending on the selected time range. This enables the admin to evaluate growth trends across different periods.

- **Staff**

  The Staff Module allows the admin to create and manage staff accounts specifically consisting of dentists and clinic assistants. It presents staff records in a table format divided into two tabs — Assistant and Dentist — enabling the admin to monitor and manage each type of staff separately based on their roles in the clinic.

  - **Create Staff Account**

    This feature allows the admin to create a new assistant or dentist account through a modal form. For an assistant account, the form requires the full name and the dentist they are assigned to. For a dentist account, the form requires the full name, specialization/s, employment type (in-house or roaming), and schedule.

  - **Staff List**

    This feature displays staff accounts in a table format with two tabs: the Assistant tab and the dentist tab, allowing the admin to monitor each type of staff separately.

    - **Staff Profile**

      Displays the staff's information as well as the list of patients they handle and the total of completed appointments. This allows the admin to have a comprehensive view of the dentist's activity and patient load within the clinic.

- **Deactivate/Reactivate Staff Account**

  This feature allows the admin to deactivate a staff account in scenarios such as termination and/or resignation and will be placed inside the deactivated accounts module, as well as reactivate it when a pre-existing account comes back.

  - **Deactivated Accounts**

    This feature displays a list of all deactivated staff accounts in the system, including their name, role, and employment type. The admin can view and reactivate any account from this list should a former staff member return to the clinic.

- **Reports**

This module lets the admin view all billing and invoice records in the system in a table format. It allows the admin to view all invoices created by the clinic.

- **Invoice Records List**

  This feature displays all the recorded payment of all appointments, including the patient ID, the time of payment, the service/s provided by the clinic, and the total value of the scheduled appointment.

- **Outstanding Balance List**

  This feature displays all the patients with outstanding balances in their account. It includes the patient ID, the date of when the last payment was made, and the amount of balance still unpaid for.

- **Filter**

  This feature allows the admin to filter records by date and service type for the Invoice Records List, and by date, amount, and patient name for the Outstanding Balance List.

Dentist's side

- **Dentist's Dashboard**

  The Dentist's Dashboard Module provides an overview of the dentist's daily activities, schedules, and assigned patients. It serves as the main working interface where the dentist can monitor appointments, manage patient flow, and access essential information needed during consultations.

  - **Daily Appointment Summary**

    This feature is presented as a set of summary cards that display the total number of today's appointments, completed consultations, and a countdown reminder for the next scheduled appointment.

  - **Today's Schedule and Patient Queue**

    This feature displays all appointments scheduled for the current day, including both scheduled and walk-in patients. It allows the dentist to monitor patient flow throughout the day by presenting information such as patient name, appointment time, type of visit, and current status, which may include waiting, in consultation, or completed.

    - **Filter**

      This feature allows appointments to be sorted based on their current status such as waiting, ongoing, or completed. It helps the dentist quickly locate and organize appointments depending on their progress in the consultation workflow.

    - **Patient Diagnosis**

      This feature allows the dentist to add diagnosis, treatment plan, and recommendations after consultation. It helps in properly recording the patient's condition and supports accurate treatment.

- **Appointment Calendar**

  This feature serves as a reference for viewing and monitoring scheduled appointments. It displays appointments in daily, weekly, and monthly views, allowing visualization of consultations across different dates. It is designed for viewing purposes only and does not allow modifications to the schedule. It includes date navigation controls for switching between calendar views and provides detailed appointment information such as patient name, time, treatment type, and status when a specific date is selected. It also highlights dates with scheduled appointments to provide a clear overview for the dentist.

- **Quick Appointment Access**

  This feature provides quick navigation through appointments labeled Past Appointments and Upcoming Appointments. Each section displays relevant appointment details such as patient name, date, time, and status for easy reference. When an appointment is selected from the patient information., the calendar automatically redirects and highlights the corresponding date. This allows quick access to the selected schedule without manually searching through the calendar.

- **Manage Schedule**

  The Manage Schedule Module functions differently for each dentist depending on their work setup. An in-house dentist has a predefined, consistent weekly schedule assigned by the clinic. Because availability is fixed, the schedule cannot be modified or created independently, but disabling the schedule on specific days is allowed. Roaming dentists, on the other hand, move between multiple branches and do not follow a fixed schedule. For this reason, roaming dentists are allowed to manage availability in the system, including adding, adjusting, or removing time slots, allowing for flexibility to accommodate scheduling needs.

- **Weekly/Monthly Schedule**

  In this feature, roaming dentists can select a day of the week and automatically add available time slots for all corresponding days, providing scheduling flexibility.

- **Disable Date**

  This feature allows both in-house and roaming dentists to block specific dates from being available for appointments ensuring that patients are unable to book on those days. Any dates that have confirmed appointments will only be allowed to be disabled in case of an emergency.

- **Add Slot**

  This feature allows roaming dentists to add new available time slots for a selected date or day. It is used to increase appointment availability, especially when additional working hours are opened or when dentists adjust their schedule to accommodate more patients.

- **Edit Slot**

  The roaming dentist can modify existing time slots within their fixed working hours. It is used to modify time intervals or update scheduling arrangements when necessary, ensuring that the appointment schedule remains accurate and properly aligned with the dentist's availability.

- **Remove Slot**

  In this feature, dentists can delete selected time slots from the schedule and prevent patients from booking them, ensuring that unavailable or no longer applicable time slots are removed to avoid scheduling conflicts.

Assistant side

- **Assistant's Dashboard**

The Dashboard Module provides the assistant an overview of the clinic's daily operations. The assistant can also perform actions directly from this module, such as viewing appointment and patient details, approving appointment requests, and rescheduling appointments.

- **Daily Summary Overview**

  This feature displays summary cards that present key daily statistics, including total revenue, number of new patients, completed consultations, and pending appointment requests. It also provides a quick view of the dentists who have scheduled appointments for the day, along with the corresponding appointment times, allowing the assistant to easily track which doctor is handling patients and at what time. The data is automatically updated based on the assistant's actions, such as marking appointments as completed or processing payments.

- **Appointment Monitoring**

  This feature allows the assistant to efficiently manage and monitor all patient appointment requests through a tabbed interface for better organization and faster processing of records. The module includes different tabs for organizing appointment statuses. The All tab displays all appointment records as the master list for overall monitoring. The Pending (For Approval & Rescheduling) tab contains new appointment requests that require action, either for approval or rescheduling depending on schedule availability. The Approved tab shows all appointment requests that have been successfully approved and confirmed by the assistant. The Rescheduled tab displays all appointments that have been moved to a new date or time. Lastly, the Cancelled tab contains all cancelled appointments for proper tracking and reference.

  - **Proof of Payment Viewer**

This feature allows the assistant to verify the patient's submitted proof of payment before approving an appointment. It is displayed in a modal when an appointment is selected, allowing the assistant to view uploaded receipts directly within the system. Appointments with submitted proof of payment can be viewed in the Pending tab for approval processing or in the All tab if they have not yet been approved. This ensures that no booking is confirmed without a valid payment record, helping maintain accurate and secure transaction validation.

- **Patient Queue Monitoring**

  This feature displays the list of patients scheduled for the current day and tracks their progress inside the clinic. It shows the patient's name, assigned dentist, appointment time, services to be rendered and a status badge indicating whether the patient is waiting, in consultation or completed. The assistant can manually update the patient's status using action buttons such as Reschedule, which allows the assistant to move the appointment to a different date or time in case of emergency or schedule adjustments, and Mark as Completed, which can only be applied once the consultation is finished and the patient already has a recorded payment transaction in the Billing Module. To prevent accidental updates the system includes an alert confirmation when the "Mark as Completed" button is selected. This ensures that appointment completion is properly aligned with verified billing records.

- **Billing and Payment Module**

  The Billing and Payment Module allows the assistant to manage all payment-related transactions within the clinic. This module serves as the dedicated interface for processing patient payments, tracking outstanding balances, and generating digital receipts, ensuring that all billing activities are properly documented and consistently reflected in the system.

- **Patient Billing Records Table**

  This feature displays all patient billing records in a table format, showing each patient's name, service rendered, total treatment cost, amount paid, remaining balance, payment method, and current payment status. It provides the assistant with a complete overview of all transactions at a glance, eliminating the need to switch between separate views to monitor payment activity.

  - **Process Payment**

    This feature allows the assistant to encode the services rendered during a consultation, select the corresponding payment method, and finalize the transaction directly from the billing table. When processing payment, the assistant can enter the exact amount the patient will pay for that visit — whether full or partial — with percentage shortcuts available for common installment amounts of the remaining balance. Once payment is processed, the patient’s billing record is automatically updated, including the remaining balance and payment status. This billing record serves as a reference for the Patient Queue Monitoring to determine if the patient is eligible to be marked as completed after the consultation is finished.

    - **Receipt Generation**

      This feature allows the assistant to generate a digital receipt after every completed transaction. It captures essential billing details such as the services availed, amount paid, payment method, and transaction date, ensuring that each payment is properly documented for both the patient and the clinic's records.

- **Installment Payment Tracking**

This feature supports installment-based payments from the billing table. It records the initial down payment, automatically updates the remaining balance after each visit, and displays a progress bar reflecting the current payment completion. The assistant can also view the full payment history per patient to monitor how much has been paid across multiple visits, reducing the need for manual computation and minimizing billing discrepancies.

- **Outstanding Balance Monitoring**

  This feature allows the assistant to identify patients with unpaid or partial balances directly from the billing table. Payment status is color-coded and labeled — paid, partial, unpaid, or installment — ensuring that the assistant can quickly spot and act on any pending balances without manually reviewing individual records.

- **Search and Filter**

  This feature allows the assistant to search for a specific patient by name or ID and filter billing records by payment status or service type, enabling faster navigation and retrieval of billing information during clinic operations.

- **Walk-in patient**

This module is designed to manage patients who visit the clinic without a prior account or scheduled appointment. It allows the assistant to encode patient information directly into the system, create a patient account, and record appointment details based on the patient's preferred date and time. This ensures that walk-in patients are properly accommodated and their records are immediately stored in the system. The module supports continuous clinic operations by handling on-site patients without requiring prior online registration.

- **Registration**

This feature provides a digital registration form accessible through a QR code or link, where walk-in patients enter their personal details, contact information, and treatment history. The submitted information is used to create a patient account and maintain organized digital records. Once registered, the patient can use the account to access the AMG website and view past and upcoming appointments.

- **Appointment Scheduling**

  This features can allow the assistant to schedule an appointment for the walk-in patient, including preferred date and time and also the preferred dentist.

Patient's side

- **Landing Module**

The Landing Module serves as the digital entry point of the clinic's appointment system. It provides an overview of the dental clinic, including its offered services, clinic information, mission and vision, and contact details. It also features clinic photos to give patients a clear idea of the facility and help establish trust and credibility. The module includes a clear "Appoint Now" button that directs patients to the registration and login modules allowing them to proceed with booking an appointment. The Landing Module serves as the first point of interaction between the patient and the clinic creating a professional and welcoming impression for both new and returning users.

- **Clinic Location**

  This feature displays the clinic's address through an interactive map feature to help users easily locate and navigate to the branch.

- **Register**

The Register Module allows patients to create an account by entering their personal details such as name, contact number, address, gender, birthdate, and password. After registration, patients can log in using their email and password to access their account.

- **One Time Password**

  To enhance account security, the system implements a multi-factor authentication process feature by sending a 6-digit verification code (OTP) to the user's registered email address. This has the timer for the duration of the unique verification code and the resend function. This ensures that the account is verified only once upon creation and confirms that the registered email belongs to the user.

- **Complete Account Setup**

  This module presents a series of questions regarding the patient's medical history and health conditions, ensuring that the clinic has all the necessary information on file before the patient's first appointment. It also includes an HMO account setup selection where the users are asked if they are an HMO holder through the system. If marked as an HMO holder, the user will provide relevant HMO details, such as membership information, for verification. If the user did not select the HMO, the account completion setup will continue as is without requiring the user to fill out the membership information details.

- **Patient's Dashboard**

The Patient Dashboard Module serves as the main interface for patients after logging in. In this module, patients can view their appointment details, including the date, time, and assigned dentist. Also the module provides easy access to other features such as updating personal information and navigating to the appointment scheduling module.

- **Calendar View**

  The Calendar Sub-Module allows patients to view available slots and request appointments based on the dentists' schedules. Patients can select a preferred date, time, and dentist for approval, after which the appointment status is updated and the patient is notified on their dashboard, ensuring scheduling remains organized and convenient for both patients and staff.

  - **Appointment Form**

    After choosing a preferred date for the appointment, the patient must complete the appointment form to finalize their booking information.

  - **Proof of Payment**

    The proof of payment section displays the phone number and QR code of the client's Gcash and Bank accounts for appointment fees, which cost ₱200. After submitting the appointment, the patient cannot book another appointment until the current appointment is either completed or rescheduled. This measure enables the system to prevent multiple or spam appointments.

- **Profile**

This module displays the patient’s personal information, including their medical and health conditions. Patients can update profile details and change the account password.

- **Appointment History**

  In this feature the patient will be able to monitor past appointments as well as outstanding balances.

Common Modules

These modules are used by all users of the system. They include basic features for accessing and managing their accounts.

- **Login**

  The Login Module allows the patients to log in to their accounts by entering their email and password.

  - **Forgot and Reset Password**

    This feature ensures account security by allowing patients to reset their password through email verification. A reset link is sent to the patient's registered email address, which redirects to the reset password module where a new password can be created.

- **Settings**

  This module allows users to manage their account settings, such as changing their password or updating basic preferences.

Other Modules

This module is accessible to different types of users within the system. It provides shared functionalities that allow some of the users to manage and access system features based on their assigned roles and permissions.

- **Patient Records**

  This module is accessible only for the admin, dentist, and assistant. It provides access to patient profiles, showing their ID, name, and personal information. It allows the users to view detailed profiles, check appointment histories, and filtering records by ID or last name for fast navigation.

- **Patient Information List**

  This feature displays a list of all patients in a table format with basic details such as name and ID, allowing the dentist and assistant to easily select and manage patient records.

  - **Patient Profile Viewing**

    This button allows the dentist and assistant to view complete patient information including personal details and appointment history before and during a consultation. HMO verification status is also displayed for patients who indicated being HMO holders; otherwise, it is not included in the profile.

    - **Patient Profile**

      This screen displayed after clicking the Patient Profile Viewing. It shows complete patient information, including personal details, contact information, and medical history. It also shows the patient’s treatment history and past consultations

    - **Appointment History**

      This feature shows a list of the patient’s personal information, time of appointment and type of treatment and the payment status.

    - **View Details Action**

      This feature shows the complete patient-submitted form.

      - **PDF Download**

        This feature allows the user to download the patient information form for documentation purposes.

- **Search and Filter**

  This feature allows fast searching and filtering of patient records by name or ID.

Limitations

The following are the limitations of the system, highlighting its current constraints and areas where functionality will not be covered.

- **Online Payment API Integration**

  The system does not integrate with third-party services such as insurance, dental imaging software, or online payment processors like GCash or PayMaya. Integrating these services would require additional technical setup and costs, which are beyond the scope of this study.

- **No HMO Verification Feature**

  The system does not include HMO validation. Although it can collect and store HMO information from patients, it only keeps these details for record purposes. The system does not handle HMO verification, approval, or financial transactions. All HMO-related processes will still be done manually by the clinic staff.

REVIEW OF RELATED LITERATURE/STUDIES/SYSTEMS

This chapter provides context for the proposed study by reviewing related literature, studies, and systems. The review highlights existing problems in clinic operations and the need for a centralized solution that addresses appointment scheduling, patient record management, and billing and payment tracking for the client, to justify the study's development.

Foreign Literature

Why every modern clinic needs a dental appointment management system.

According to the article of Bartolome, J (2025), Many dental clinics face challenges with scheduling, leading to inefficiencies and potential revenue loss. An article from Synergy Advantage explores how implementing a dental appointment management system can address these challenges, offering benefits for both the clinic and its patients. One of the benefits of having an appointment management system in a dental clinic is that patients find it convenient for them because the system provides easy and flexible online scheduling, automated reminders from the system reduces no-shows appointment schedules by keeping patients informed regarding their scheduled appointment, online appointment also provide the clinics with real-time data tracking to help scheduling strategies, optimize operational procedures and enhance the clinics popularity .

Dental technology: Then and now

According to the Greater Manchester Institute of Technology (n.d.), innovations such as artificial intelligence (AI), digital dental X-rays, and computer-aided design and manufacturing (CAD/CAM) systems have transformed the way dental services are delivered by improving diagnostic accuracy, enhancing treatment planning, and increasing efficiency in clinical procedures. These technologies allow dental professionals to provide more precise and consistent patient care. However, the integration of such digital systems also requires clinics to adapt by developing technical skills, updating workflows, and keeping up with continuous technological changes to meet modern healthcare demands.

These two literatures are connected to the proposed project for the client, as both emphasize the importance of adopting digital systems to improve efficiency and service quality. Similar to what the foreign studies highlighted, the client currently faces challenges in scheduling, record management, and billing due to its manual operations. The proposed Online Appointment System aims to address these issues by introducing computer-based features such as automated scheduling, organized record management, and real-time monitoring, allowing the clinic to transition toward a more efficient, accurate, and modernized workflow.

Local Literature

Health dept implements appointments system in public hospitals.

According to Inquirer.net (2025), the Philippine Department of Health (DOH) implemented a new Patient Appointment System (PAS) to address long queues and inefficiencies in public hospitals. Initially rolled out in three DOH-run hospitals — Jose R. Reyes Memorial Medical Center, San Lazaro Hospital, and Dr. Jose Fabella Memorial Hospital — the system enables patients to book consultations online in just two to three minutes. For those without internet access, hospital personnel assist them at in-hospital kiosks, ensuring wider accessibility.

This initiative reflects a national-level recognition of the problems caused by manual scheduling and patient flow, mirroring the challenges faced by the client. By referring to this PAS launch, the proposed online appointment system is aligned with broader healthcare modernization trends in the Philippines, reinforcing that digital scheduling can improve not just accessibility, but overall service delivery in dental care.

Foreign Studies

The use of various appointment systems among patients visiting academic outpatient centers in Kerman and the evaluation of patients' perspective and satisfaction

The study of Bagheri et al. (2022) conducted on 332 patients in academic outpatient centers in Kerman to evaluate their satisfaction with different types of appointment systems. The results showed that the satisfaction level was only moderate, with a score of 49.12 out of 100. Among the systems used, patients found the web-based application more convenient compared to the USSD and IVR systems. However, the study also mentioned that even with digital systems, clinics still face challenges in integration and usability, which can affect patient satisfaction.

Stakeholders' experiences, perceptions and satisfaction with an electronic appointment system: A qualitative content analysis

Likewise, Nabovati et al. (2025) examined the experiences and satisfaction of stakeholders who used an electronic appointment system in a specialty clinic in Iran during 2022–2023. The study found that while the system made scheduling easier and more convenient, there were still several issues such as unclear requirements, poor system management, and technical difficulties that affected its performance and reliability.

Both studies show that even though appointment systems make scheduling more efficient, they still need to be properly managed and designed to meet the needs of both staff and patients. These findings are important for the client's situation because they highlight the importance of having a reliable, user-friendly, and well-integrated appointment system. With these insights, the proposed project can help the client develop a more efficient system that prevents the same problems found in other healthcare clinics.

Local Studies

An Implementation and Evaluation of Web-Based Appointment System for the Mindanao State University – Main Campus

The study conducted by Ampuan and Delena (2022) proposed a web based appointment system due to the lack of an effective appointment system in the Office of the Mindanao State University in Marawi City, They found that even the busiest in the university faced issues such as wasting time in scheduling and delayed tasks. With the start of pandemic the need for a reliable scheduling platform became more critical, their proposed system was evaluated using the System Usability Scale and Technology Acceptance Model obtained a 90.2% satisfactory rate proving that the improving scheduling system makes efficiency and less waiting time for the client.

This study is relevant to the client because the clinic currently handles appointments through multiple platforms such as Messenger, SMS, and manual recording, which leads to delays, unorganized scheduling, and longer waiting time for patients. Similar to the issues identified in the study, the clinic experiences difficulty in managing appointment requests efficiently, especially during busy hours. By implementing a web-based appointment system, patients can book appointments in advance, reducing waiting time and improving the overall flow of scheduling. The system can also organize appointment data, provide a clear schedule for the staff, and minimize manual errors in recording. This will allow the clinic staff to manage appointments more efficiently while providing faster and more convenient service to patients, improving both operational performance and patient satisfaction.

Foreign Systems

Investigating patient use and experience of online appointment booking in primary care: Mixed methods study.

The system investigated by Atherton et al. (2024) is an online appointment-booking platform designed to improve accessibility and ease of scheduling for primary care services, allowing patients to self-book, reschedule or cancel appointments and receive confirmation via digital means. Its scope includes supporting a broad patient base, reducing administrative burden on clinics, and enhancing patient experience through convenience. However, the system has notable limitations: it depends on digital access and literacy so patients without internet or comfort with online tools may be excluded), integration with legacy healthcare systems may be weak, and there are concerns regarding verifying non-standard documentation such as foreign patient IDs, plus potential language/ cultural barriers and regulatory issues when dealing with international or foreign documentation.

Patient Opportunities to Self-Schedule in a Large Multisite, Multispecialty Medical Practice: Program Description and Uptake of 7 Unique Processes for Patients to Successfully Self-Schedule (and Reschedule) Their Medical Appointments

North et al. (2024) studied patient self-scheduling systems in medical institutions and found that managing multiple scheduling methods across different specialties can be complex. They identified seven self-scheduling approaches and mapped five primary and two secondary processes. The study showed that self-scheduling improves convenience by letting patients view real-time availability before choosing a time slot. After selecting a schedule, patients input their personal details and concerns, following a calendar-first process that makes appointment booking more efficient and user-friendly.

The studies of Atherton et al. (2024) and North et al. (2024) show that online self-scheduling appointment systems improve healthcare efficiency and patient convenience by allowing users to view availability, book, and manage appointments independently, but they also present limitations such as digital literacy barriers, system integration challenges, and difficulties in managing complex administrative requirements. These findings are important to the proposed system because they highlight the need for a simple, accessible, and secure appointment platform that ensures efficient scheduling and proper handling of patient information. Guided by these studies, the proposed Online Appointment System aims to improve workflow efficiency, enhance accessibility, and strengthen coordination between patients and clinic staff while addressing common issues found in existing systems.

Local Systems

An Information System for Private Dental Clinic with Integration of Chat-Bot System: A Project Development Plan. International Journal of Advanced Trends in Computer Science and Engineering

The study of Alejandrino and Pajota (2023), focused on designing a Web-based Dental Information Management System for Romero Dental Clinic. Its scope is managing patient dental records, scheduling with the help of a rule-based chatbot tracking inventory for dental supplies and notifying the patients through SMS. On the other hand the study is limited to its design and focused only on selected modules for the clinic, it did not cover full development with other external systems.

UP PGH Online Appointment System (OAS)

The Philippine General Hospital (PGH), under the University of the Philippines Manila, implemented an Online Appointment System (OAS) to improve patient access to medical services and reduce overcrowding in hospital facilities. Through this system, patients can access an online portal where they first select their desired clinic or department and view available appointment schedules. Patients are then able to choose their preferred date and time slot before proceeding to complete the required personal and medical information for appointment submission.

The study of Alejandrino and Pajota (2023) presents a web-based dental information management system for Romero Dental Clinic that integrates patient record management, appointment scheduling supported by a rule-based chatbot, inventory tracking of dental supplies, and SMS notifications; however, its scope is limited to system design and selected modules only, without full implementation or integration with external systems. Similarly, the UP PGH Online Appointment System (OAS) allows patients to select a clinic or department, view available schedules, and then choose their preferred date and time before filling out personal and medical details for appointment submission. These systems are relevant to the proponents' proposed project for the client as they demonstrate how dental and hospital information systems can streamline scheduling, record-keeping, and related administrative processes, providing useful insights for possible feature enhancements and future system expansion.

Synthesis

The reviewed literature, studies, and systems show that manual and fragmented processes in healthcare and dental clinics lead to inefficiencies such as scheduling conflicts, delayed services, and errors in record and billing management. These issues are also present in the client's situation, where multiple platforms and manual methods result in disorganized workflows and time-consuming processes. While digital appointment systems improve efficiency and patient experience, their effectiveness depends on proper integration and usability. Anchored on these findings, the proposed Online Appointment System aims to address these problems by integrating scheduling, patient records, and billing into a single, organized platform.

METHODOLOGY

The development of the web-based Online Appointment System for the client will follow the **Agile methodology.** Agile is a project management approach that focuses on breaking down the development process into smaller, manageable tasks that can be completed in short iterations. This allows continuous improvement, flexibility, and adaptation to changes based on user feedback. Agile promotes collaboration among team members and stakeholders, enabling faster delivery of functional system components while ensuring that the system meets the client's needs effectively.

The project will specifically utilize the **Kanban framework**, where tasks are represented as cards on a visual board and organized into columns based on their progress. Team members move tasks from backlog to completion, allowing better workflow visualization, efficient task management, and continuous delivery. This approach helps teams limit work in progress, identify bottlenecks, and improve overall efficiency (Asana, 2023). *(To see the figure please go to Appendix B, Page 70 & 71).*

Planning & Requirements

During this step in the iterative process, the team will define the project plan and align on the overall project objectives. This is the stage where the team will outline any hard requirements, things that must happen in order for the project to succeed. This detailed analysis will help the team to decide whether or not a project is feasible before commencing work. (J. Martins, 2025)

Analysis & Design

This phase focuses on analyzing the business needs and technical requirements of the system. The team organizes and refines the information gathered during the planning stage to create a clear system design. This includes structuring system features, defining workflows, and preparing the foundation for development using flowcharts, wireframing like figma and database designing. (J. Martins, 2025)

Implementation

During this phase is where the development team, particularly the programmers, begin creating the actual system based on the approved design and analysis. Using the blueprint developed during the design stage, such as wireframes created in Figma, the team translates these visual and structural plans into a working system. The functionalities, user interface, and system features are coded and integrated according to the specified requirements. This stage focuses on turning the planned design into a functional initial version of the system, ensuring that it aligns with the intended workflow and objectives of the project. (J. Martins, 2025)

Testing

After the implementation phase, the system undergoes testing to evaluate its functionality and overall performance, including usability testing, system testing, and user feedback collection to ensure it meets user requirements. In this phase, the team will test the developed system based on the guidelines provided in the article, where the programmers will verify if all features are working correctly, identify errors or bugs, and ensure that the system aligns with the design and intended objectives, while making necessary improvements before proceeding to the next stage. (J. Martins, 2025)

Evaluation & Review

In the final phase, the team evaluates the results of the testing process to determine necessary improvements and refinements in the system. In this stage, the team reviews the feedback, identified issues, and overall system performance to ensure that it meets the desired quality and project objectives. If revisions are needed, the programmers and developers will return to the analysis and design phase to improve the system based on the findings. This iterative process continues until the system becomes fully functional, reliable, and aligned with the client's requirements, with the duration of each cycle depending on the complexity of the project. (J. Martins, 2025)

Technical Background

This section provides an analysis of the client's existing technological infrastructure, followed by a detailed examination of the proposed integrated technologies outlined in the proposal. At present, the client's daily operations uses both manual and semi-digital tools to manage their operations which can lead to inefficiencies and data inconsistencies. The following discussion presents the client's detailed current technological operations and how the proposed system integrates these processes into a more efficient and accessible platform.

- **Overview of Current Tools to be Used**

  Daily Appointment Monitoring and Patient Inquiry Handling Process

  Currently, the client manages appointment scheduling using Google Calendar while communicating with patients through Viber, Messenger, and SMS. At the beginning of the day, assistants check scheduled patients and prepare for procedures. If there are no scheduled patients, assistants respond to inquiries through messaging platforms *(see Appendix E, p. 80 - 83, No. 1 & 12)*. Since all assistants handle scheduling, overlapping responsibilities sometimes cause double booking, especially when one assistant updates an appointment without the other being aware *(see Appendix E, p. 80 - 89, No. 1 & 26.2)*.

  Patient Appointment Booking and Information Collection Process

  Patients request appointments through calls or messages, and assistants collect basic details such as name, contact number, concern, and availability *(see*

Appendix E, p. 80 - 82, No. 2 , 5 & 7). For HMO patients, additional information such as HMO details, date of birth, and company is collected for verification before scheduling *(see Appendix E, p. 81 - 82, No. 5 - 7)*. HMO verification usually takes 1–2 days but may extend to weeks depending on the HMO provider, which can lead to loss of patients while waiting *(see Appendix E, p. 81 & 92 No. 6 & 31).* Once verified, appointments are placed in Google Calendar using color-coded labels for scheduling status *(see Appendix E, p. 82 & 91, No. 8 & 30).* Appointment reminders are manually sent via SMS or calls before the appointment date *(see Appendix E, p. 83 & 93, No. 11 & 33).*

Patient Record Management and Storage Process

Patient records are stored in paper-based mini booklets containing contact details, medical history, procedures, and consent forms *(see Appendix E, p. 84, No. 14).* These are arranged alphabetically in filing cabinets, with each letter assigned to a specific drawer *(see Appendix E, p. 93, No. 34)*. Because records are permanently stored, more cabinets are continuously added, which creates space issues over time *(see Appendix E, p. 96, No. 40).* When a booklet becomes full, continuation sheets are attached to extend the record *(see Appendix E, p. 97, No. 44)*.

Patient Record Retrieval Process

When a patient arrives, assistants manually retrieve their records from filing cabinets. Even if arranged alphabetically, searching still takes time due to similar surnames and large volume of files *(see Appendix E, p. 84 & 94, No. 16 & 36).* Records are pulled one by one or sometimes in advance, but this can cause delays if multiple patients are scheduled *(see Appendix E, p. 94, No. 37)*. Handling multiple booklets at the same time sometimes leads to misplacement or incorrect filing *(see Appendix E, p. 95, No. 37.1 & 39).*

Billing and Payment Tracking Process

After procedures, dentists provide billing details while assistants manually record payments in the patient booklet and issue receipts *(see Appendix E, p. 85 & 100, No. 18 & 51)*. Payment records are then encoded in Google Sheets for monitoring *(see Appendix E, p. 101 - 102, No. 56 & 58)*. For long-term treatments like braces, balances are tracked manually and updated every visit as payments are made *(see Appendix E, p. 98 - 99, No. 46 & 48)*. At the end of the day, records in the booklet and Google Sheets are compared to ensure accuracy *(see Appendix E, p. 85, No. 19)*. Manual processing sometimes leads to discrepancies such as missed entries or incorrect computations *(see Appendix E, p. 101, No. 57)*.

Proposed System

In the proposed system, patients will no longer need to contact the clinic manually through chat applications. Instead, they can directly access the clinic's official web-based appointment platform, starting with the Landing Page, where the clinic's information and services are displayed. From there, patients can register or log in to their accounts and proceed to the Appointment Scheduling Module.

The system includes a built-in calendar interface that displays available time slots based on the dentists' schedules. Patients can select their preferred dentist, date, and time, and submit an appointment request. Once submitted, the secretary receives a notification through the dashboard, where the request can be reviewed, approved, or rescheduled if necessary. This centralized scheduling process addresses the clinic's issue of double booking and miscommunication caused by handling appointments across multiple messaging platforms and manual calendar updates. By allowing all appointment requests to pass through one unified scheduling module, the system improves coordination, reduces scheduling conflicts, and minimizes delays in confirming appointments.

After approval, dentists can view their assigned patients for the day and record diagnostics, procedures performed, and treatment details directly into the system. This replaces the clinic's paper-based patient records, solving the problem of slow record retrieval, misplaced files, and damaged booklets. With digital patient records stored in a searchable database, staff can quickly access patient histories, update records in real time, and securely maintain information without relying on physical filing cabinets. This improves efficiency while ensuring that patient records remain organized and protected.

Once the consultation or treatment has been finished, the patient then proceeds to the billing process, which is managed by the secretary. In the proposed system, the secretary is able to mark the appointment as completed inside the billing module, where they can input the services provided within the appointment which the system automatically provides the total amount to be paid. For cash payments, the secretary inputs the amount given by the patient, and the system automatically generates the change for convenience, while for cashless transactions such as GCash, the secretary inputs the reference number as proof of payment. Any outstanding balances can also be recorded and monitored within the system. This directly addresses the clinic's current billing issues, where partial payments and balances are manually tracked using paper notes and spreadsheets, leading to delayed updates and payment discrepancies. By automating billing calculations and payment balance monitoring, the system improves billing accuracy and ensures that patient payment records are updated consistently.

This integration ensures that appointment scheduling, patient records, and billing processes are centralized within a single system. By replacing disorganized tools such as separate messaging platforms, Google Calendar, Google Sheets, and paper-based records, the proposed system reduces manual errors, improves transparency, enhances data security, and increases operational efficiency. It manages appointments more efficiently without switching between different tools. It also makes it easier for the staff to access and update information whenever needed. By directly resolving the clinic's challenges in scheduling, record management, and billing, the proposed system provides a faster and more reliable workflow, resulting in improved service efficiency and a more seamless experience for both staff and patients.

- **Calendar of Activities**

This section provides a detailed sequence of activities that the proponents would undergo in completing the project, covering each phase from the start of development up to its completion.

PHASE 1: Planning & Requirements

WEEK 1: January 22 – January 28, 2026

During this week, the proponents forwarded the previous version of the Transcript of Interview (TOI) and previous study report to the project adviser for review. Initial consultation was conducted to discuss the project direction and documentation requirements.

WEEK 2: January 29 – February 4, 2026

During this week, the proponents continued interviewing the client to gather detailed information about the clinic's existing appointment process, to compose a new Transcript of Interview (TOI) together with the necessary supporting documents as required by the adviser.

WEEK 3: February 5 – February 11, 2026

During this week, the proponents finalized the newly Transcript of Interview (TOI) and completed the client interview. The adviser instructed the team to review the thesis template and begin drafting Chapter 1, particularly identifying the general problem and the main problem of the study.

WEEK 4: February 12 – February 18, 2026

During this week, the proponents drafted Chapter 1 of the study, which covers the Project Context, Purpose and Description, Objectives, and the Scope and Limitations The team underwent a preliminary examination and received comments for improvement. Revisions were made accordingly, including the addition of proper citations, refinement of the Review of Related Literature, and clarification of the Project Context and Purpose and Description to improve coherence and avoid redundancy.

WEEK 5: February 19 – February 25, 2026

During this week, the proponents added necessary citations and revised the chapter 2 Review of Related Literature (RRL) based on the adviser's comments. The Project Context, Purpose and Description were further refined for clarity and consistency. The documentation was prepared for submission and approval to proceed to Chapter 3.

WEEK 6: February 26 – March 4, 2026

During this week, the proponents revised and continued the creation of chapter 3 with further instructions from the adviser. The Methodology has been changed and refined, Resources has been simplified, Requirements Analysis has been refined, Requirements Documents has been created, Design of Software, System, Product, and/or Process has been refined. Reference has been updated with its new reference links and old reference links removed.

WEEK 7: March 5, 2026 - March 11, 2026

During this week, the proponents focused on refining the project documentation based on the adviser’s feedback and the formatting guidelines from the library. The team completed the necessary forms, revised the Scope section by restructuring it into user types, and ensured that the document followed the required format. Unnecessary white spaces were removed to improve the overall layout and readability making the document paper easier to read and more organized. Additionally, the proponents created the initial Gantt Chart to visualize the project timeline and align it with the documented activities.

WEEK 8: March 12, 2026 - March 18, 2026

During this week, the proponents continued enhancing the documentation by organizing and improving the Review of Related Literature (RRL), including adding appropriate titles and aligning the synthesis with the study. The team also worked on developing the Requirements Documentation and began the Design of Software, System, Product, and/or Processes. This includes outlining the introduction, planning, designing, implementation, and deployment aspects of the proposed system. These activities helped establish a clearer structure for the system development phase.

WEEK 9: March 19, 2026 - March 25, 2026

During this week, the proponents finished attaching titles to each Review of Related Literature, Study, and System, connecting the Synthesis more on the study and refining the Requirements Documentation and Design of Software. The team will proceed to improving the Objectives and clarify more on what features each module will have.

WEEK 10 : March 26, 2026 - April 1, 2026

During this week, the proponents refined the overall format of the documentation including white spaces, cut-out titles, and grammatical errors. The proponents discussed the current paper and summarized the ideas regarding the individual parts of the documentation.

WEEK 11: April 2, 2026 - April 8, 2026

During this week, the proponents reformatted the scope and instead listed all features of individual modules as well as adding the common and other modules. The proponents also changed the methodology into Agile while using Kanban as its framework.

WEEK 12: April 9, 2026 - April 15, 2026

During this week, the proponents added 2 additional supporting RRLs regarding the relevance of creating an appointment through selecting a preferred schedule first in the interactive calendar. The gantt chart is revised into including the expected activities the proponents will do in the future.

WEEK 13: April 16, 2026 - April 22, 2026

During this week, the proponents will complete their final revision of the documentation and this will also mark the completion of Capstone 1— the first phase of the development of the system.

PHASE 2: Analysis & Design

WEEK 14 - 21: April 23, 2026 - June 17, 2026

The proponents will gather and finalize system requirements, define user roles and permissions, and produce system models such as use case diagrams, ERDs, and data flow diagrams. UI mockups and wireframes will be created in Figma, the database schema will be designed, and all requirement decisions will be documented.

PHASE 3: Implementation

WEEK 22 - 38: June 18, 2026 - October 14, 2026

The proponents will set up the development environment and build all four modules — authentication, appointment scheduling, patient records, and billing — alongside the administrator dashboard. The frontend will be integrated with the Django backend and PostgreSQL database, with unit testing conducted alongside each module as development progresses.

PHASE 4: Testing

WEEK 39 - 42: October 15, 2026 - November 11, 2026

The proponents will conduct the system integration testing and user acceptance testing across all user roles, validating that the system meets its defined requirements. Identified bugs will be resolved and all test cases and results will be formally documented.

PHASE 5: Evaluation & Review

WEEK 43 - 44: November 12, 2026 - November 25, 2026

The proponents will analyze user acceptance testing feedback, finalize system and thesis documentation, and prepare the final defense presentation.

Gantt Chart

A gantt chart is commonly used in project management to show each activity when it was done and when it will be done. Each activity represented by a bar the position and length of the bar reflects the start date, duration and end date of the activity. (*What Is a Gantt Chart? Gantt Chart Software, Information, and History*, n.d.-b) *(see Appendix C, page 73)*.

- **Resources**

This section identifies the specific hardware and software resources required for the successful development, deployment, and operation of the proposed Online Appointment System for the client. These resources ensure that the system performs efficiently and remains accessible to both the clinic staff and patients.

Software Resources:

- **Figma**

  Will be used as a cloud-based collaborative design tool for creating flowcharts, interface mockups, and system designs. It will help the developers visualize the user flow and overall structure before building the system. It will also allow easy collaboration among team members, enabling them to share ideas and design prototypes.

- **Visual Studio Code**

  Will be used as the primary Integrated Development Environment (IDE) for writing and managing all project code. It will support multiple programming languages such as HTML, CSS, JavaScript, Python, and Django through its extensions, making development faster and more organized. It will be used as the main platform for developing the system.

- **Github**

  Will be used for version control and team collaboration. It will allow the team to track changes, manage different versions of the code, review updates, and work together without overwriting each other's contributions. It will ensure safe storage of the codebase and make development more organized.

- **PostgreSQL**

  Will be used as the system’s Database Management System (DBMS). It will store all patient information, appointment schedules, billing data, and system records in a secure and structured database. It will also allow efficient data retrieval and updating through queries, ensuring that information can be accessed quickly when needed. Additionally, it will support multiple users accessing the system at the same time while maintaining data accuracy and consistency.

- **Google Docs**

Will be used for documentation writing and collaborative editing. It will allow the team to prepare the manuscript, track revisions, and organize the study content efficiently.

Hardware Resources:

Minimum Requirements:

- **Processor: Intel Core i3 (10th Gen) or AMD Ryzen 3**
  Will be used as the minimum processing unit to run the system efficiently. It will handle basic tasks such as loading the web application, processing user inputs, and managing simple operations without lag.

- **RAM: 8 GB**
  Will provide enough memory for smooth multitasking while running the system. It will allow users to open the application, browser tabs, and other necessary tools without performance issues.

- **Storage: 256 GB SSD**
  Will be used for storing the system files, database, and other necessary data. The SSD will ensure faster boot time, quicker data access, and overall improved system responsiveness compared to traditional hard drives.

- **Graphics: Integrated Graphics**
  Will be used for rendering the system's user interface. Since the system will be web-based and does not require heavy graphics processing, integrated graphics are sufficient for smooth operation.

- **Operating System: Windows 10 or Windows 11**
  Will serve as the platform where the system runs. This will

provide compatibility with development tools, browsers, and system dependencies required for proper functionality.

Requirement Analysis

The requirements analysis for the client is presented in this section. To ensure that the developers must integrate the system's essential components in order to guarantee that the system successfully integrates the requirements and expectations of the intended beneficiaries. To further define the system requirements, the proposed system is analyzed in terms of the following aspects:

Who

The proposed system is intended for the users of the client. This includes patients who are the primary recipients of the dental clinic services who will be able to schedule book appointments online and provide all necessary information, dentists who provide clinical services and access treatment information, and dental assistants who handle patient appointments, manage patients record and process billing transactions.

What

The proposed system will function as an online appointment system for the client that will be serve as the core platform to manage the appointment scheduling, patient records, and billing processes. The function for appointment scheduling this will be display available slots for dentist and prevent double booking for each dentist. Patients record will organize contact information, medical history, and treatment records for quick and secure access for every patients. Billing and payment processes will be automated and tracking balances for every patients. This will be help to the client to enhance their clinic operations, reduce error and improve the quality of services.

Where

The proposed system will be used at AMG Dental Clinic: Commonwealth Branch where the system will be installed as well as online. Dentists and dental assistants will use the system at the clinic using desktop computers or any types of gadgets, while patients will be able to use the system at any location where the internet is available. The system will provide a central platform to run all the operations of the clinic using a single online system instead of using several physical forms, records, and communication applications.

When

The proposed system will be available during the clinic's operating hours, from Monday to Saturday which is 10:00 AM to 4:00 PM, for internal operations such as updating records, confirming appointments and processing payments. However, the appointment feature may be accessible at any time allowing patients to schedule appointments even outside clinic hours. The system will be used before appointments for booking and verification, during consultations for accessing and updating patient records and after procedures for billing and payment documentation.

How

The current procedures of the clients relies on manual and separated platforms, Appointment Scheduling is handled through different applications such as Viber, Messenger and SMS, while the schedules is tracked by using Google Calendar. Patients record maintained manually in mini booklets and stored in different cabinets making the retrieval time consuming and prone to misplacement. The billing informations is recorder to the paper and then record again into google sheets, thus increasing the risk of error and inconsistencies. The proposed system will be streamlined these scattered processes by digitizing appointment scheduling, patients record and integrating billing into a single centralized website platform.

Requirements Documentation

Based on the identified operational inefficiencies of the client, the proposed system is required to fulfill the following functional specifications across four core modules. The user authentication and access control module must grant secure, credential-protected access to patients, assistants, dentists, and administrators, ensuring each is limited to functions relevant to their role. Administrators must have overall access to all modules and must be able to create and deactivate user accounts across all roles. The appointment scheduling module must allow patients to view real-time available slots and book appointments online, while assistants can confirm, reschedule, or cancel bookings through the same platform. The patient records management module must maintain a digital profile for each patient storing their contact details, medical and dental history, and procedure records, accessible to both assistants and dentists during clinic operations. For patients who identify as HMO holders during account creation, their profile will include an HMO status field reflecting either a verified or pending state, while profiles of non-HMO patients will not carry this field. The billing and payment tracking module must record payments, track full and partial balances, log payment methods, store proof of cashless transactions, and link all billing entries to the corresponding patient profile for end-of-day reconciliation.

Design of Software, System, Product, and/or Processes

The proposed system will be a web-based platform designed to centralize and streamline the daily operations of the client. The system will be developed following the Agile methodology, allowing the proponents to iteratively design, test, and refine each module throughout the development process.

Planning

The planning phase will begin with the proponents analyzing the client's operational needs to establish system requirements that directly address the identified gaps in scheduling, records management, and billing. Flowcharts will also be produced during this phase to map out user interactions and data flow before development begins.

Designing

Using Figma, the proponents will develop wireframes and high-fidelity mockups tailored to each user role — keeping the patient interface simple and intuitive, organizing the assistant and dentist interfaces for efficient clinical workflows, and providing the administrator with broader access covering monitoring, reporting, and account management. UML diagrams will also be produced, including a Use Case Diagram, Sequence Diagram, and Class Diagram to define system interactions, process flows, and structural relationships, alongside an Entity-Relationship Diagram to define the data structure and relationships managed by PostgreSQL.

Implementation

The system will be built using HTML, CSS, and JavaScript for the frontend, Python with the Django framework for the backend, and PostgreSQL as the database. Role-based access control will be implemented to restrict each user to permitted functions, and the four core modules will be developed and integrated incrementally following the Agile approach.

Deployment

Once all core modules have been fully implemented and tested, the system will be made available to the client for actual use in their daily clinic operations

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APPENDICES

APPENDIX A. RESEARCH PERSONS

APPENDIX A. RESEARCH PERSON

Name: Dr. Macaisa, Michelle Abada
Position: Owner/Dentist of AMG Dental Clinic
Affiliation: Client

Name: Macaisa, Czarina Ricci
Position: Secretary of AMG Dental Clinic
Affiliation: Client

Name: Dr. Macaisa, Andrea Natassja
Position: Dentist of AMG Dental Clinic
Affiliation: Client

APPENDIX B. RELEVANT IMAGES / TABLES

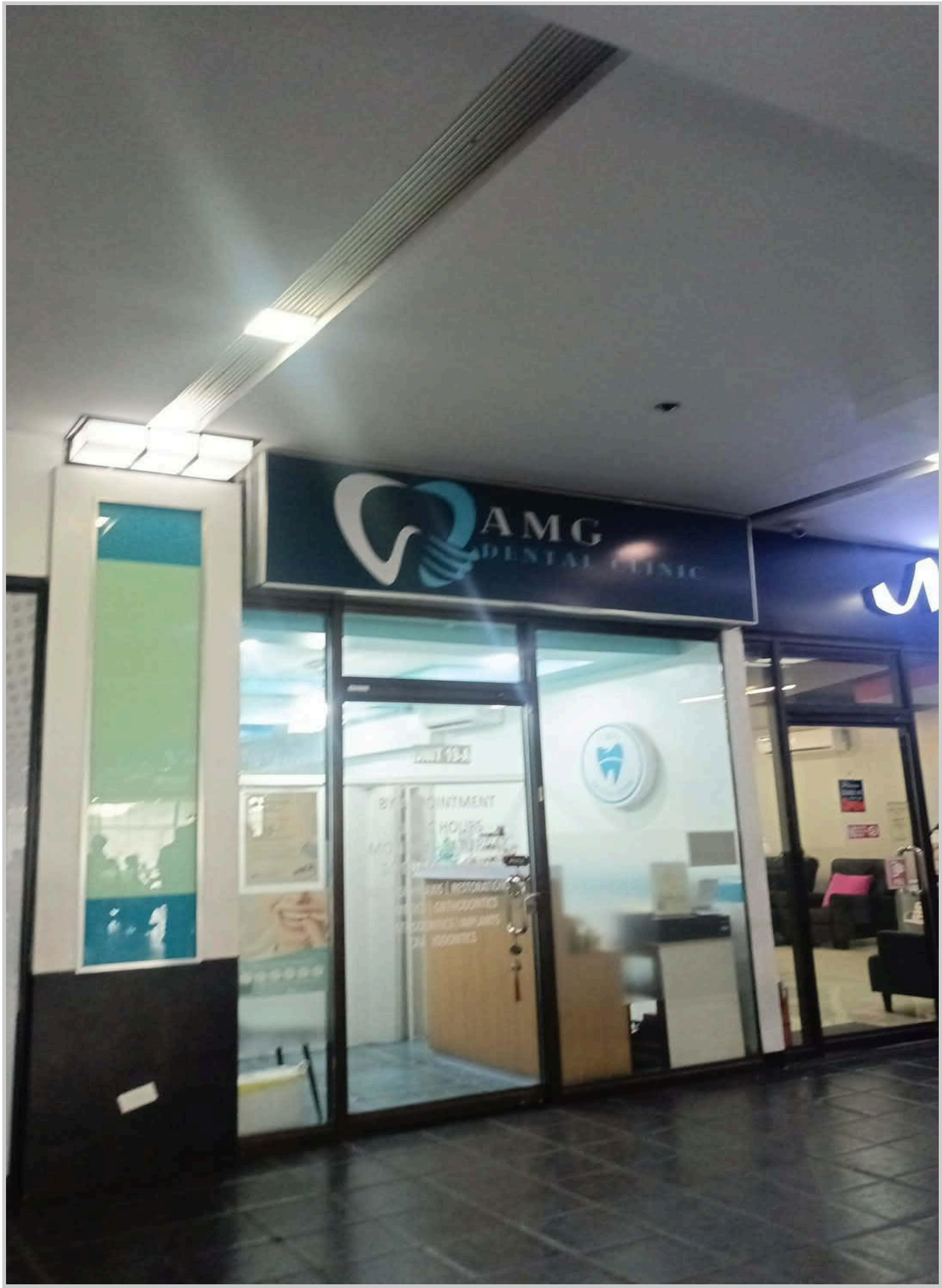


Image 1: Commonwealth Branch.

This image presents the entrance of the clinic, showing the main access point for patients entering the facility.

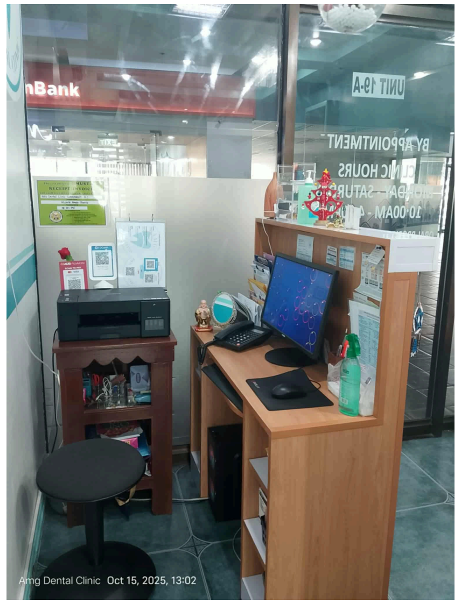


Image 2: Front Desk Area

This image presents the clinic's front desk area, along with the dentist and the corresponding assistant, illustrating their roles during daily operations.

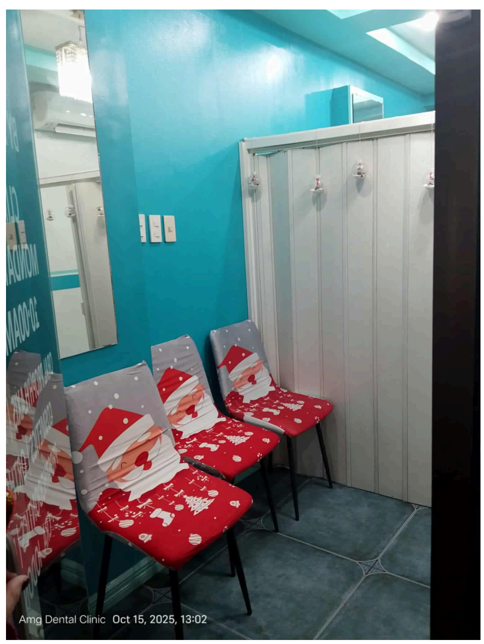


Image 3: Waiting Area

This image presents where patients wait before their scheduled appointment or treatment.

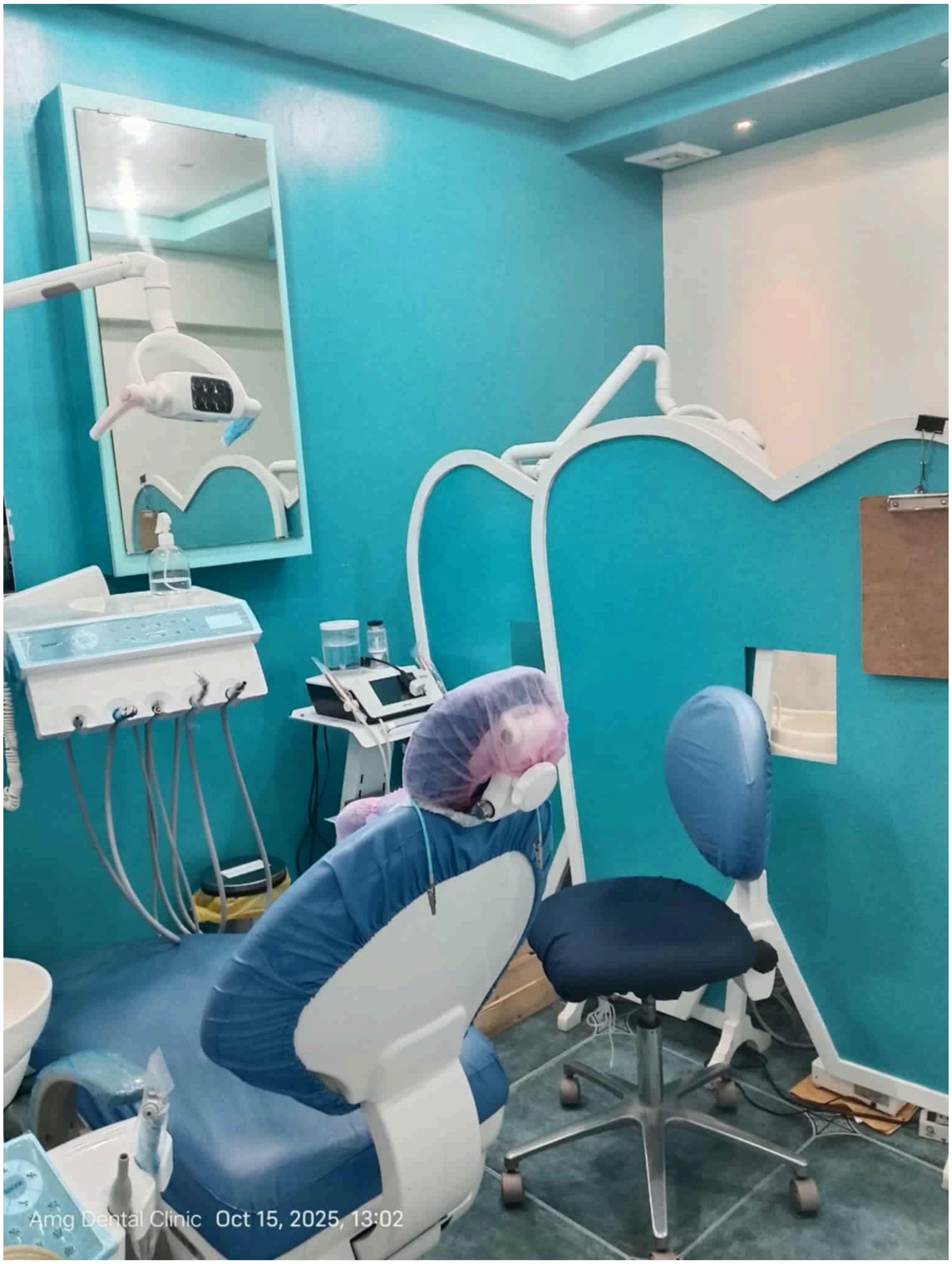


Image 4: Dental Chair

This image presents the chair used for patient examinations and dental procedures in the clinic.

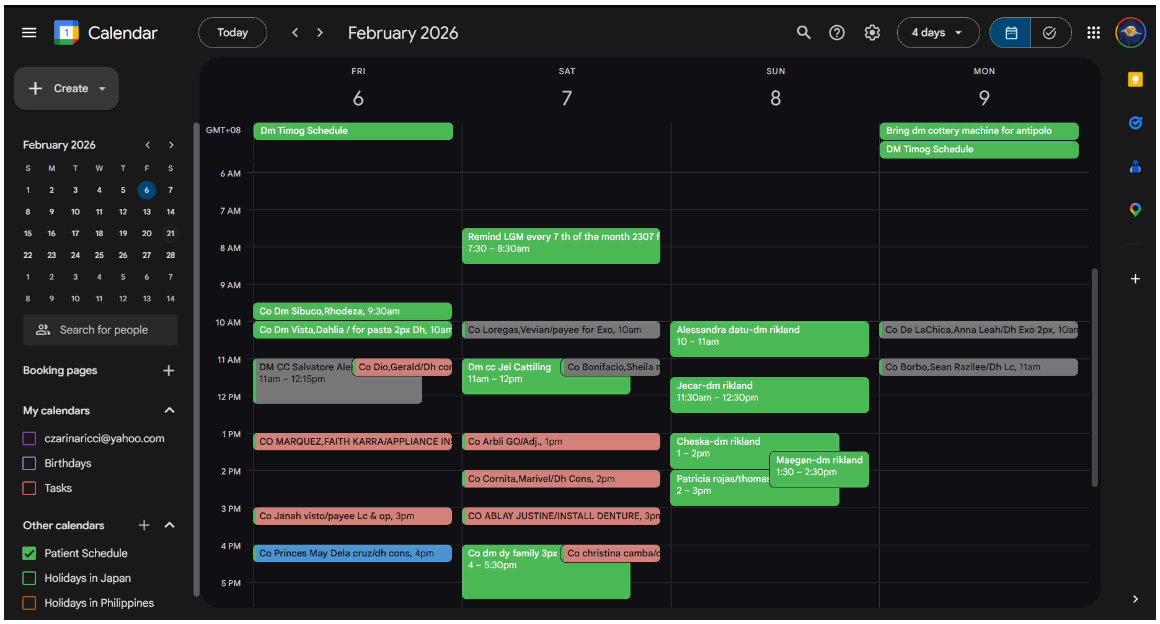
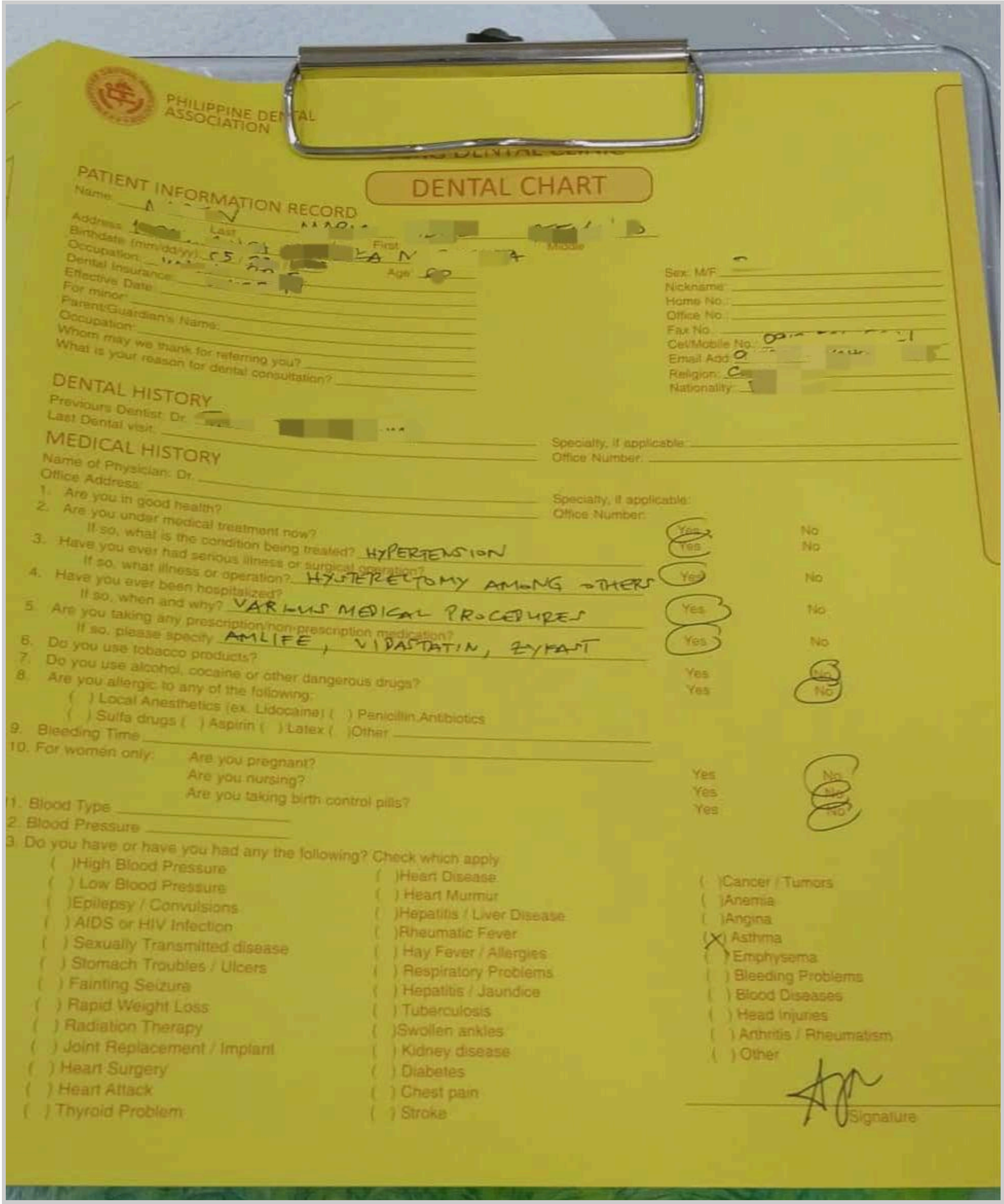


Image 5: Google Calendar

This image shows the Google Calendar used for scheduling patient appointments, where colors indicate status: pink for confirmed, grey for cancelled, and blue for unconfirmed appointments.


PHILIPPINE DENTAL ASSOCIATION

DENTAL CHART

PATIENT INFORMATION RECORD

Name: Last First Middle

Address:

Birthdate (mm/dd/yy):

Occupation:

Dental Insurance:

Effective Date:

For minor:

Parent/Guardian's Name:

Occupation:

Whom may we thank for referring you?

What is your reason for dental consultation?

Sex: M/F

Nickname:

Home No.:

Office No.:

Fax No.:

Cel/Mobile No.:

Email Add:

Religion:

Nationality:

DENTAL HISTORY

Previous Dentist: Dr.

Last Dental visit:

Specialty, if applicable:

Office Number:

MEDICAL HISTORY

Name of Physician: Dr.

Office Address:

Specialty, if applicable:

Office Number:

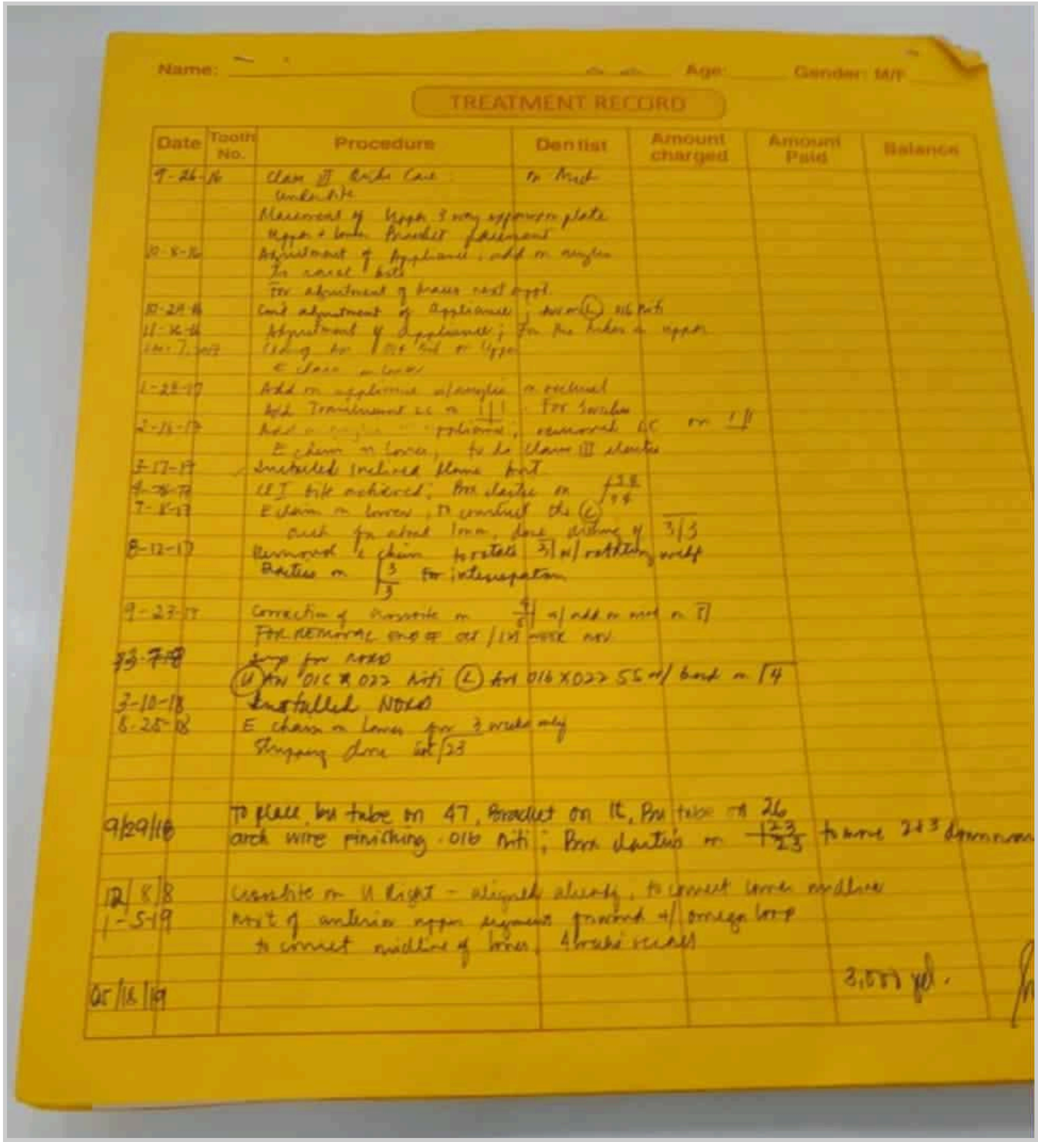
1. Are you in good health? Yes No
2. Are you under medical treatment now? Yes No
   If so, what is the condition being treated? HYPERTENSION
3. Have you ever had serious illness or surgical operation? Yes No
   If so, what illness or operation? HYSTERECTOMY AMONG OTHERS
4. Have you ever been hospitalized? Yes No
   If so, when and why? VARIUS MEDICAL PROCEDURES
5. Are you taking any prescription/non-prescription medication? Yes No
   If so, please specify AMLIFE, VIDASTATIN, ZYKAST
6. Do you use tobacco products? Yes No
7. Do you use alcohol, cocaine or other dangerous drugs? Yes No
8. Are you allergic to any of the following:
   ( ) Local Anesthetics (ex. Lidocaine) ( ) Penicillin Antibiotics
   ( ) Sulfa drugs ( ) Aspirin ( ) Latex ( ) Other
9. Bleeding Time
10. For women only: Are you pregnant? Yes No
    Are you nursing? Yes No
    Are you taking birth control pills? Yes No
11. Blood Type
12. Blood Pressure
13. Do you have or have you had any the following? Check which apply

( ) High Blood Pressure
( ) Low Blood Pressure
( ) Epilepsy / Convulsions
( ) AIDS or HIV Infection
( ) Sexually Transmitted disease
( ) Stomach Troubles / Ulcers
( ) Fainting Seizure
( ) Rapid Weight Loss
( ) Radiation Therapy
( ) Joint Replacement / Implant
( ) Heart Surgery
( ) Heart Attack
( ) Thyroid Problem
( ) Heart Disease
( ) Heart Murmur
( ) Hepatitis / Liver Disease
( ) Rheumatic Fever
( ) Hay Fever / Allergies
( ) Respiratory Problems
( ) Hepatitis / Jaundice
( ) Tuberculosis
( ) Swollen ankles
( ) Kidney disease
( ) Diabetes
( ) Chest pain
( ) Stroke
( ) Cancer / Tumors
( ) Anemia
( ) Angina
(X) Asthma
( ) Emphysema
( ) Bleeding Problems
( ) Blood Diseases
( ) Head Injuries
( ) Arthritis / Rheumatism
( ) Other

Signature

Image 6: Patient Record Booklet(Front)

This image shows the front cover of the patient record booklet, containing the patient's basic identification details.



Name: Age Gender: M/F

TREATMENT RECORD

Date	Tooth No.	Procedure	Dentist	Amount charged	Amount Paid	Balance

Image 6.1: Patient Record Booklet(Back)

This image shows the Treatment Record section of the patient booklet, where procedures and payments are recorded, including notes for outstanding balances.

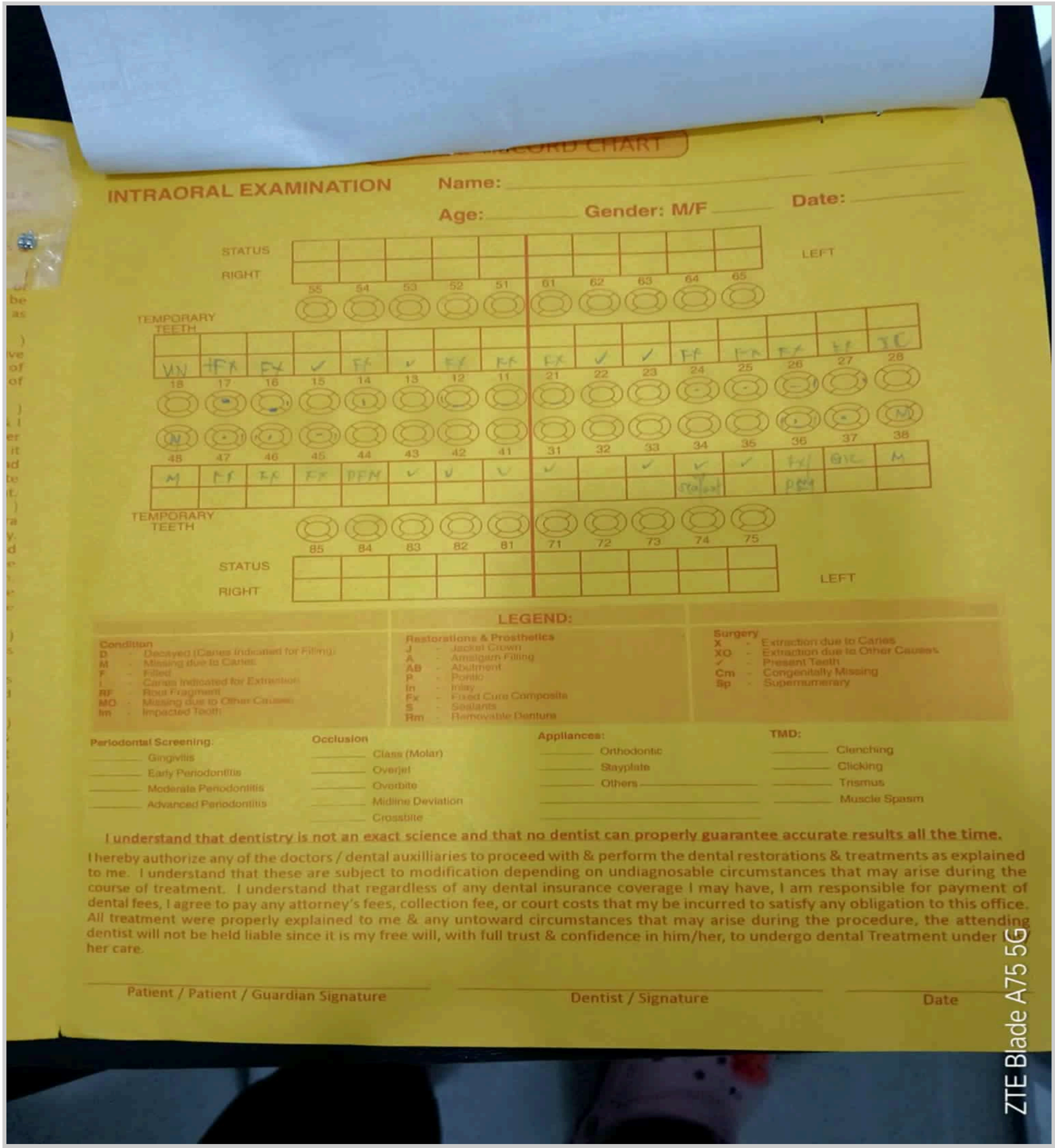

INTRAORAL EXAMINATION
Name:
Age: Gender: M/F Date:
STATUS
RIGHT
LEFT
TEMPORARY TEETH
LEGEND:
Periodontal Screening
Occlusion
Appliances:
TMD:
I understand that dentistry is not an exact science and that no dentist can properly guarantee accurate results all the time.
I hereby authorize any of the doctors / dental auxilliaries to proceed with & perform the dental restorations & treatments as explained to me. I understand that these are subject to modification depending on undiagnosable circumstances that may arise during the course of treatment. I understand that regardless of any dental insurance coverage I may have, I am responsible for payment of dental fees, I agree to pay any attorney's fees, collection fee, or court costs that my be incurred to satisfy any obligation to this office. All treatment were properly explained to me & any untoward circumstances that may arise during the procedure, the attending dentist will not be held liable since it is my free will, with full trust & confidence in him/her, to undergo dental Treatment under his/ her care.
Patient / Patient / Guardian Signature
Dentist / Signature
Date

Image 6.2: Patient Record Booklet(Inside-right page)

This shows the dental intraoral examination chart used to record the condition of each tooth, including damages, restorations, and other dental findings.

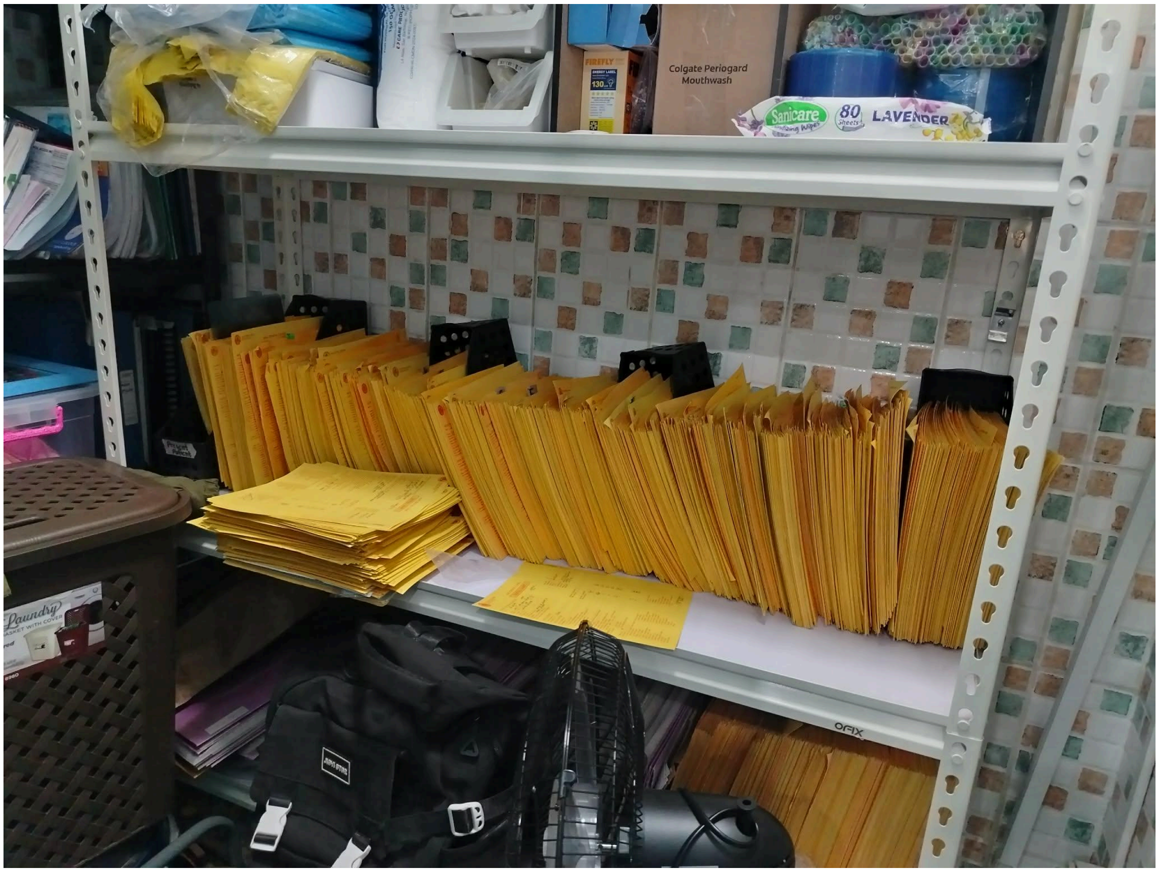


Image 7: Patient Record and Filling Cabinets

This image shows the filing cabinets used in the dental clinic where patient records are stored.

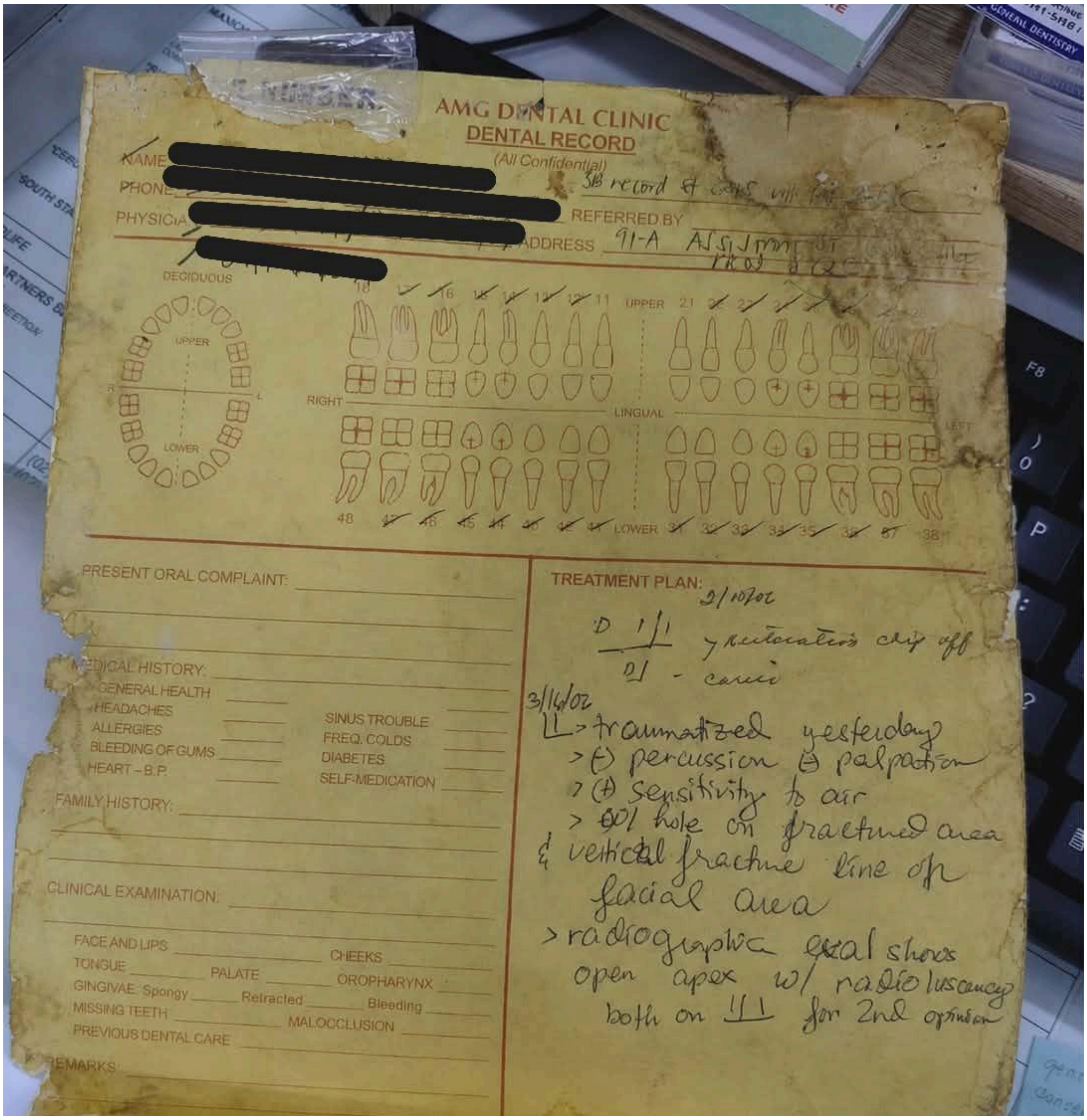
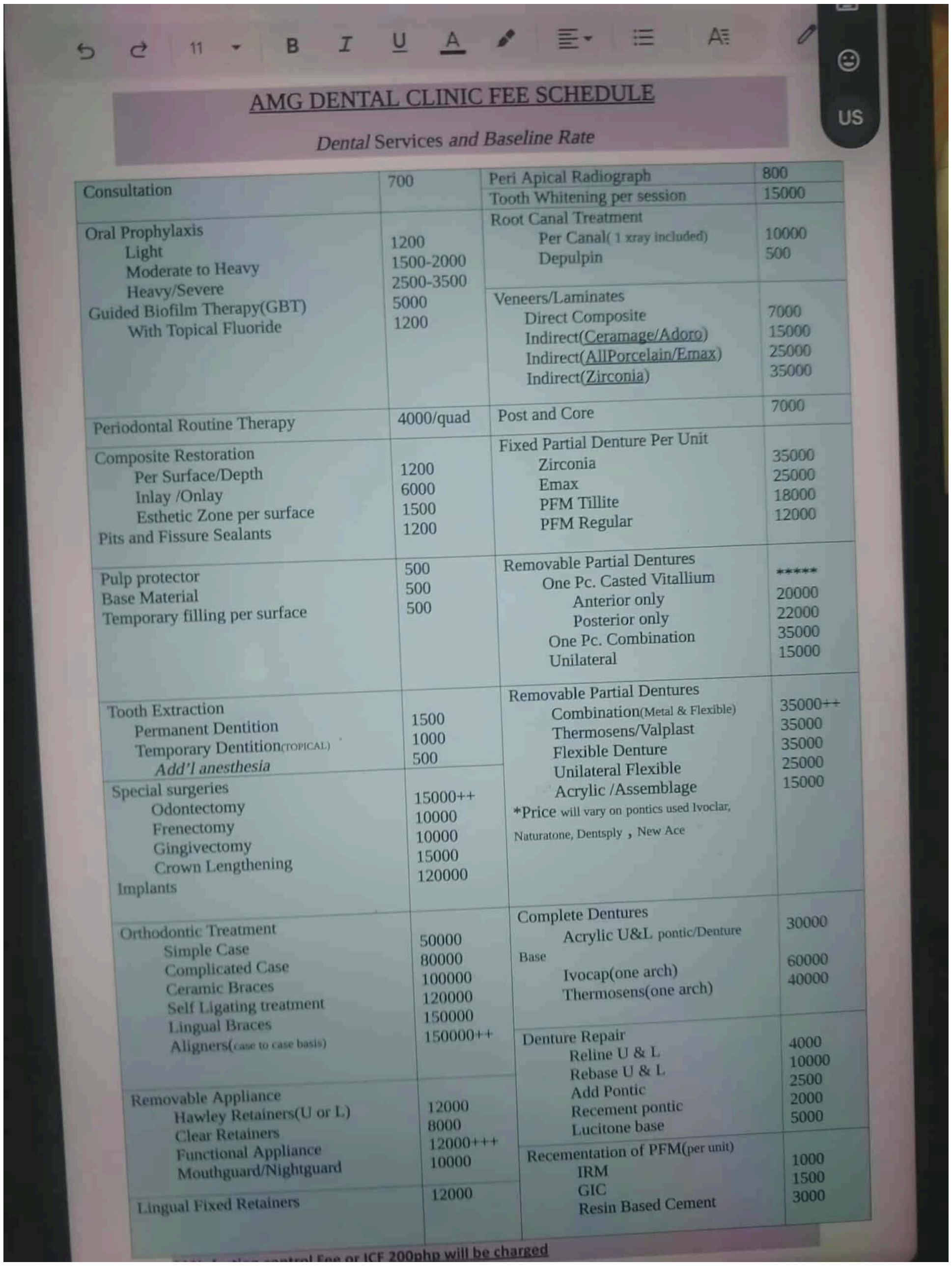


Image 8: Old Damaged Patient Record

This image shows an old and damaged patient record used by the dental clinic, where important clinical notes and patient details are written manually. Over time, the record has deteriorated due to wear, tearing, and possible exposure to moisture, making some information difficult to read and increasing the risk of data loss.



AMG DENTAL CLINIC FEE SCHEDULE

Dental Services *and Baseline Rate*

Consultation	700
Oral Prophylaxis	
Light	1200
Moderate to Heavy	1500-2000
Heavy/Severe	2500-3500
Guided Biofilm Therapy(GBT)	5000
With Topical Fluoride	1200
Periodontal Routine Therapy	4000/quad
Composite Restoration	
Per Surface/Depth	1200
Inlay /Onlay	6000
Esthetic Zone per surface	1500
Pits and Fissure Sealants	1200
Pulp protector	500
Base Material	500
Temporary filling per surface	500
Tooth Extraction	
Permanent Dentition	1500
Temporary Dentition(TOPICAL)	1000
Add'l anesthesia	500
Special surgeries	
Odontectomy	15000++
Frenectomy	10000
Gingivectomy	10000
Crown Lengthening	15000
Implants	120000
Orthodontic Treatment	
Simple Case	50000
Complicated Case	80000
Ceramic Braces	100000
Self Ligating treatment	120000
Lingual Braces	150000
Aligners(case to case basis)	150000++
Removable Appliance	
Hawley Retainers(U or L)	12000
Clear Retainers	8000
Functional Appliance	12000+++
Mouthguard/Nightguard	10000
Lingual Fixed Retainers	12000

Peri Apical Radiograph	800
Tooth Whitening per session	15000
Root Canal Treatment	
Per Canal(1 xray included)	10000
Depulpin	500
Veneers/Laminates	
Direct Composite	7000
Indirect(Ceramage/Adoro)	15000
Indirect(AllPorcelain/Emax)	25000
Indirect(Zirconia)	35000
Post and Core	7000
Fixed Partial Denture Per Unit	
Zirconia	35000
Emax	25000
PFM Tillite	18000
PFM Regular	12000
Removable Partial Dentures	
One Pc. Casted Vitallium	*****
Anterior only	20000
Posterior only	22000
One Pc. Combination	35000
Unilateral	15000
Removable Partial Dentures	
Combination(Metal & Flexible)	35000++
Thermosens/Valplast	35000
Flexible Denture	35000
Unilateral Flexible	25000
Acrylic /Assemblage	15000
*Price will vary on pontics used Ivoclar, Naturatone, Dentsply , New Ace	
Complete Dentures	
Acrylic U&L pontic/Denture Base	30000
Ivocap(one arch)	60000
Thermosens(one arch)	40000
Denture Repair	
Reline U & L	4000
Rebase U & L	10000
Add Pontic	2500
Recement pontic	2000
Lucitone base	5000
Recementation of PFM(per unit)	
IRM	1000
GIC	1500
Resin Based Cement	3000

Fee or ICF 200php will be charged

Image 9: Base Prices of Treatment Plan

This Image shows the base prices of each treatment the dentist can provide for the patient.

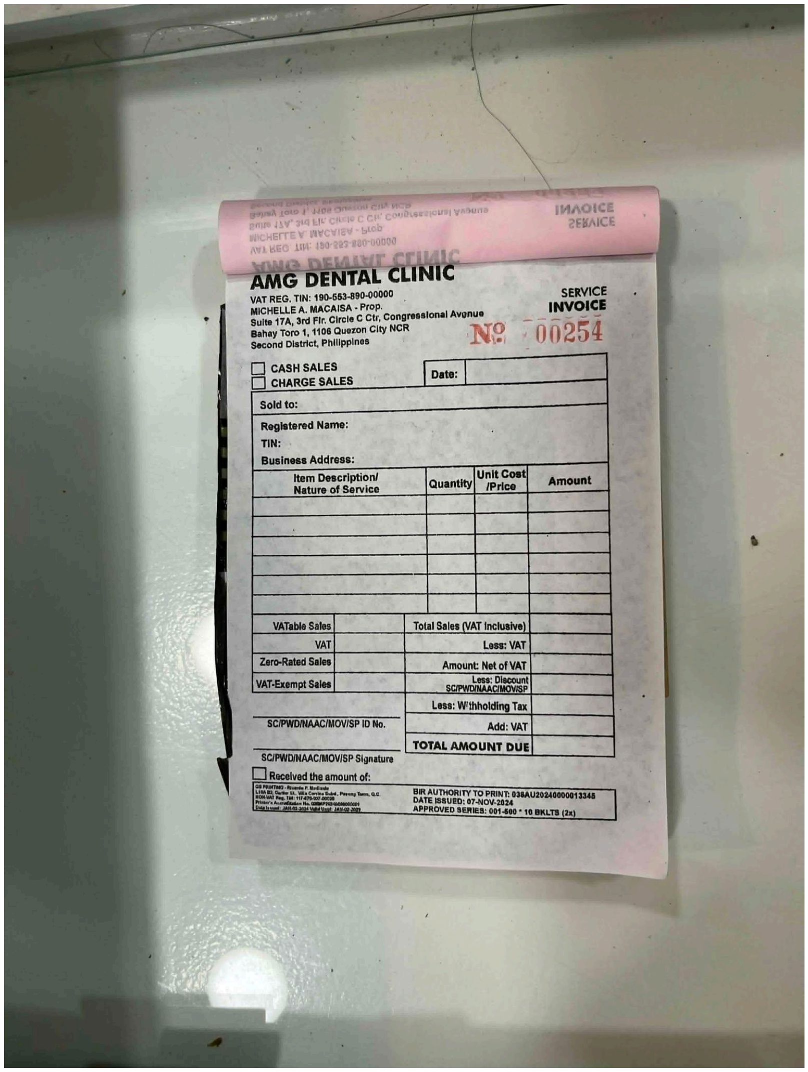


Image 10: Dental Clinic Receipt

This shows the official receipt used in the dental clinic to document patient payments and transaction details.

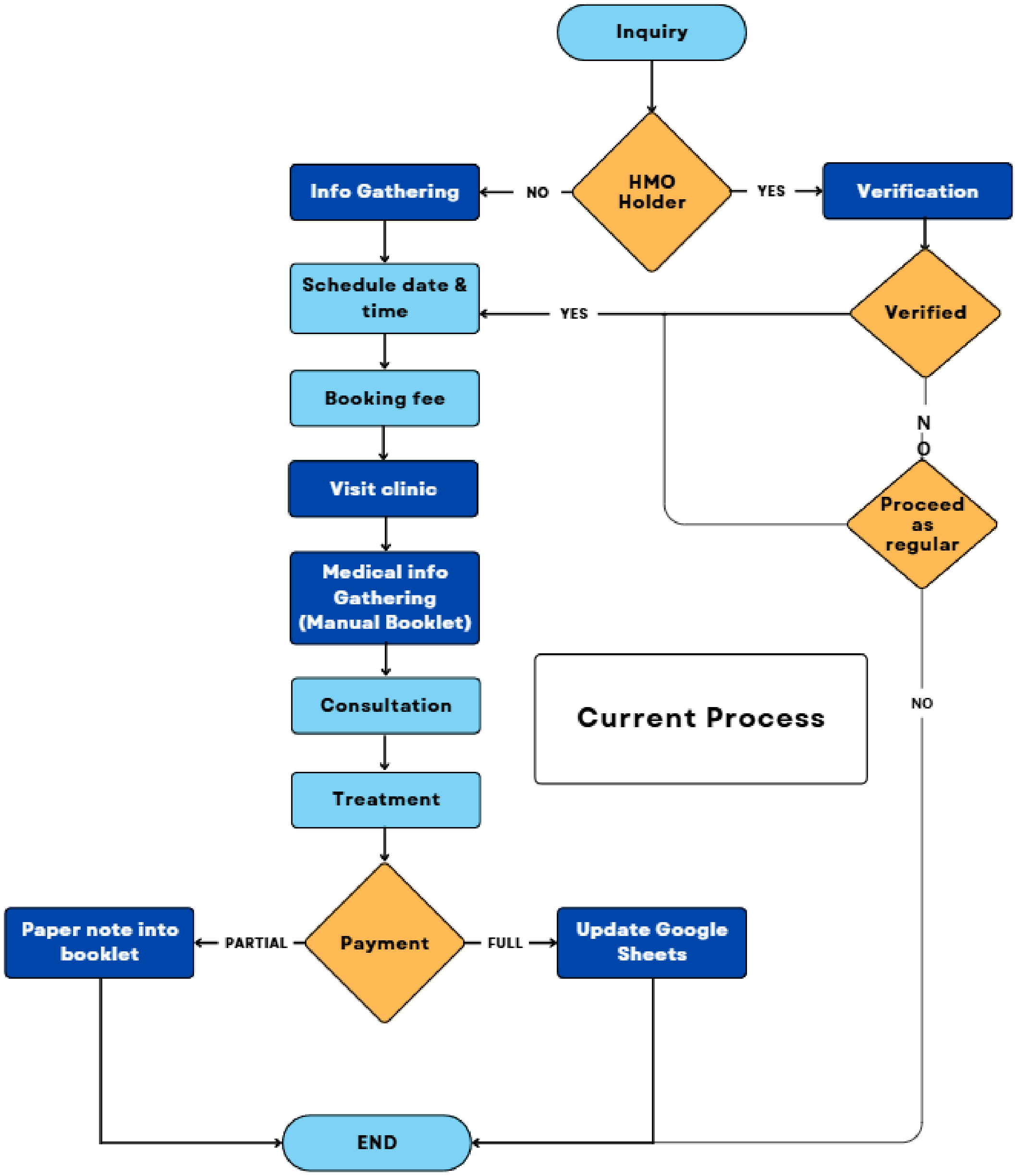


Figure 1: Current Process of the Client

This flowchart illustrates the clinic's current workflow, from patient inquiry and appointment scheduling to record management and billing. It shows the manual processes involved in handling daily clinic operations.

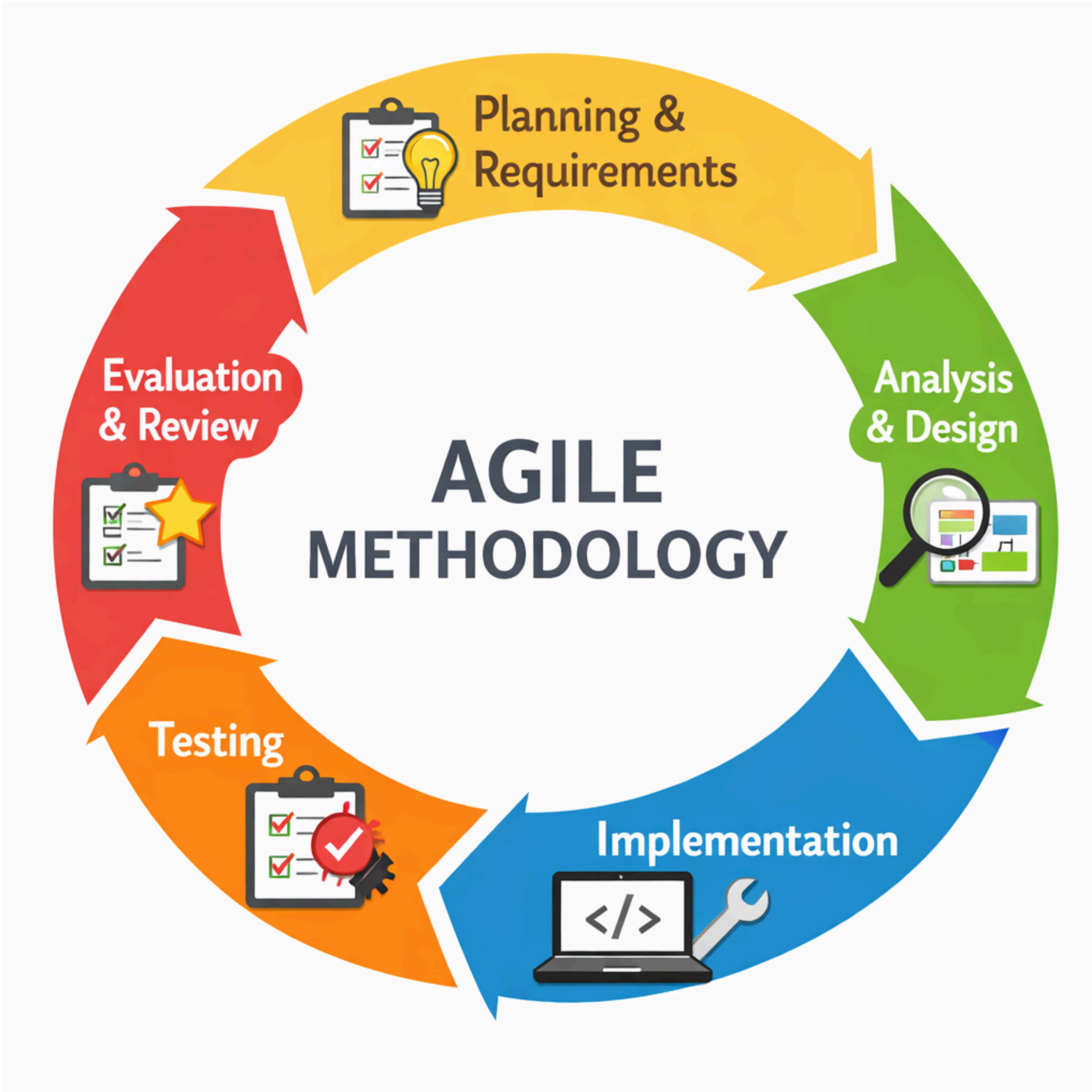


Figure 2: Agile Methodology Diagram

This figure shows the iteration of the Agile process used in the study, where development is divided into cycles of planning, design, implementation, testing, and review. Each cycle allows continuous improvement and ensures the system adapts to the client's needs.

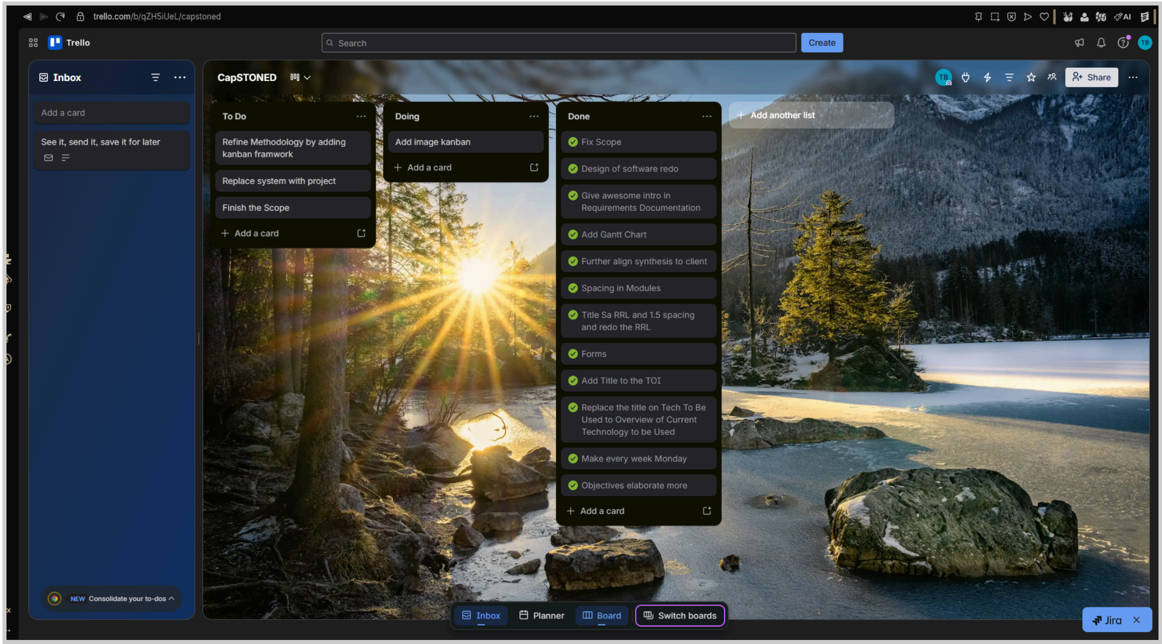


Figure 3 : Kanban Framework Diagram

This figure illustrates the Kanban framework used in the study, where tasks are organized into columns based on their progress. It helps visualize workflow, manage tasks efficiently, and ensure continuous development.

APPENDIX C. GANTT CHART

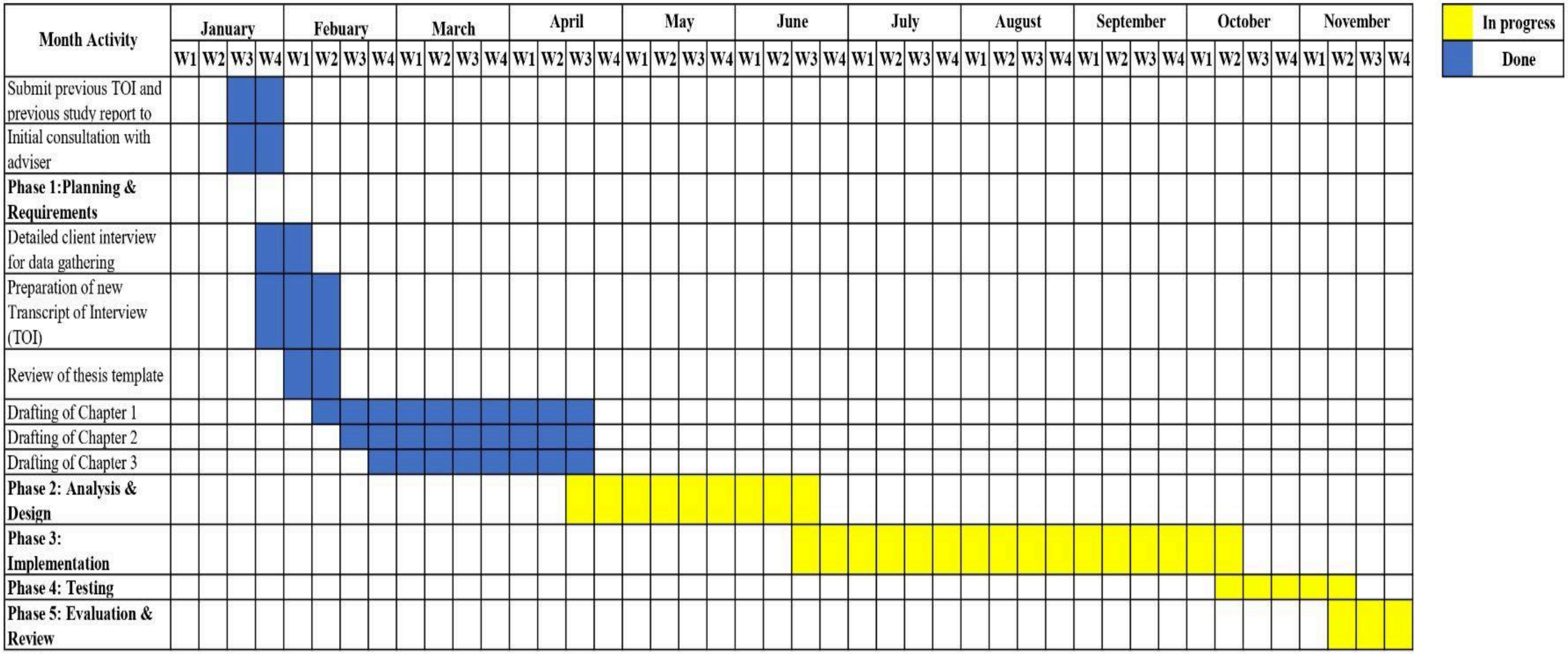


Table 1 : Gantt Chart of Activities

This table presents the project timeline and schedule of activities for the development of the system. It shows the sequence of accomplished tasks, their duration, and the expected time frame for each phase, helping to monitor progress and ensure that all activities are completed within the planned schedule.

APPENDIX D. INTERVIEWEE QUESTIONNAIRE

FIRST QUESTIONNAIRE

Question 1 : Who is the owner of the business and what does the owner do on a day-to-day basis?

Question 2: Briefly describe a typical day at your clinic, focusing on how appointments are scheduled, recorded, and managed from opening to closing.

Question 3: Approximately how many patients do you serve per day or week?

Question 4: How many people work at the clinic, and what does each person handle day-to-day?

Question 5: Are there different categories of patients (e.g., HMO, walk-in, emergency)? If yes, does scheduling or recording their appointments require different steps?

Question 6: Can you describe what happens from the moment a patient first contacts you for an appointment until they actually come in?

Question 7: What patient information do you collect when booking appointments, and where does this information go?

Question 8: How do you record confirmed appointments? What details do you write down or input?

Question 9: How do you handle appointment cancellations and rescheduling? What's that process like?

Question 10: How do you handle walk-in patients or dental emergencies when your schedule is already full?

Question 11: Do you send appointment reminders to patients? If yes, how?

Question 12: What challenges do you face when responding to patient inquiries or appointment requests?

Question 13: What tools, apps, or systems do you currently use to manage appointments, records, and billing?

Question 14: How are patient records currently stored? Walk me through where information lives and how it's organized.

Question 15: What information is typically included in a patient's record?

Question 16: When you need to find a patient's record, what's that process like? Who does it, and how long does it take?

Question 17: Have you experienced situations where records were hard to find, incomplete, or had conflicting information? Can you give an example?

Question 18: After a patient finishes their treatment, what happens next with billing and payment? Do all patients pay in full right away?

Question 19: When you're checking that all payments for the day were recorded, what's that process like? How do you catch errors or missing entries?

Question 20: What problems do you run into with billing or payment tracking?

Question 21: What are the three biggest problems related to managing appointments, patient information, or payments that you deal with repeatedly?

Question 22: If we built a system for your clinic, what features or capabilities would be most valuable to you?

Question 23: Is there anything about managing the clinic day-to-day that we haven't talked about yet that causes frustration or takes up a lot of time?

SECOND QUESTIONNAIRE

Question 24: Last time you mentioned that assistants handle appointment scheduling, patient records, and billing and payments. With two assistants working at the same time, how do you divide those responsibilities — do you each handle all three, or do you each take on specific tasks?

Question 25: From the moment a patient first messages you to the moment their appointment is fully confirmed, how long does that usually take—and what causes delays?

Question 26: How much of your day goes into managing appointments? Does it affect assisting during procedures?

Question 27: Have you ever encountered patients switching communication channels during booking and causing confusion?

Question 28: Can you explain what each calendar color represents and how it works day to day?

Question 29: Why does HMO verification take 1 to 2 days or longer?

Question 30: Have you encountered cases where patient information did not match HMO details and caused issues?

Question 31: Are appointment reminders done manually or automatically?

Question 32: How many filing cabinets are used and how are they organized?

Question 33: Are there cases where records are damaged, lost, or exposed to water?

Question 34: You mentioned it takes 5–10 minutes to find a record. Why does it take that long?

Question 35: Do you pull records one by one or in bulk before patients arrive?

Question 36: Are there cases where records are not returned properly to their drawer?

Question 37: Can you give an example of a misplaced record during a patient visit?

Question 38: Have you experienced running out of storage space for records?

Question 39: Are all family members recorded separately as patients?

Question 40: How do you ensure patient booklets are fully completed after procedures?

Question 41: Who fills in patient records after procedures?

Question 42: How do you handle records when booklets become full?

Question 43: How do you ensure both booklets are retrieved together for patients with multiple records?

Question 44: How do you track long-term treatment progress across multiple visits?

Question 45: How long does it take to set up a new patient's record before treatment starts?

Question 46: How do you track outstanding balances for long-term treatments?

Question 47: Is there a required down payment percentage for treatments?

Question 48: Is the remaining balance divided across treatment duration?

Question 49: How do you make receipts?

Question 50: Do you encounter problems when making receipts?

Question 51: Has a patient ever requested to pay less than the required down payment?

Question 52: What happens if a patient disappears or does not return?

Question 53: Have there been discrepancies in Google Sheets records among staff?

Question 54: How do you track payments and procedures in Google Sheets?

Question 55: How do payment recording discrepancies happen?

APPENDIX E. TRANSCRIPT OF INTERVIEW

FIRST INTERVIEW

<u>Czarina Ricci Macaisa 11/16/25</u>

Date & Signature Over Printed Name

Position: Assistant

1. Who is the owner of the business and what does the owner do on a day-to-day basis?

Assistant: Dr. Michelle is the owner of the clinic, and she also manages a lot of the daily activities. She sees the patients herself, does dental procedures like consultations, extractions, and other treatments. She handles a lot of the admin work too. She personally transacts with clients for the procurement of equipment, tools, materials, and other dental necessities for the clinic. She manages the bank accounts and personally files the taxes for BIR. She also handles employee salaries and ensures all permits for health regulation compliance are in order to keep the clinic running. She is the one who applies for and decides on HMO affiliations as well.

2. Briefly describe a typical day at your clinic, focusing on how appointments are scheduled, recorded, and managed from opening to closing

Assistant: After opening the clinic, setting up the instruments and equipment, assistants check the scheduled patients and prepare for each procedure. If there are no scheduled patients, assistants will answer inquiries through social media and sms for possible patients. If there are scheduled patients, assistants will help dentists during the procedure then clean up after the procedures. Before closing the clinic, instruments and equipment should be cleaned and sterilized for the following day.

3. Approximately how many patients do you serve per day or week?

Assistant: A single dentist can accommodate 6-8 patients per day, that's 12-16 patients in total if the second dentist is available. A dentist can handle 4-5 patients for more complex procedures that take 1-2 hours each.

4. How many people work at the clinic, and what does each person handle day-to-day?

4.1. Does anyone handle multiple responsibilities—like both patient scheduling and billing?

Assistant: Each dentist has an assistant. Any dentist can be assisted by any assistant and vice versa but the clinic usually has the same number of assistants and dentists. Billing and scheduling falls to the assistants so does the cleaning up and preparing of patients and instruments. Dentists also access the calendar and patient records when needed.

5. Are there different categories of patients (e.g., HMO, walk-in, emergency)? If yes, Does scheduling or recording their appointments require different steps?

5.1 Probe: Do any of these patient types require you to do things differently—like different forms, verification steps, or scheduling rules?

Assistant: Yes. The clinic also accommodates HMO patients. HMO patients have a different scheduling since they require verification. Non-HMO patients can schedule right away compared to HMO patients that need to wait for the verification before getting an appointment. The verification process takes about 1 to 2 days.

6. Can you describe what happens from the moment a patient first contacts you for an appointment until they actually come in?

Assistant: Patients either message or call us. We ask for their concern and for their availability. They would need to leave us their name and contact number for their booking appointment. HMO patients on the other hand need to be verified first before they get an appointment schedule. Patients cannot choose their own schedule directly — we offer available time slots based on the shared calendar.

7. What patient information do you collect when booking appointments, and where does this information go?

Assistant: Non-HMO patients are asked for names and their contact number. HMO patients are asked for their HMO details and contact number. For HMO patients specifically, we request their full name, contact number, date of birth, HMO card number, and company for verification purposes. The information is saved on their appointment schedule since the schedule or calendar is digitalized. We also require a ₱200 fee for infection control before booking an appointment.

8. How do you record confirmed appointments? What details do you write down or input?

Assistant: There are different colors on the calendar used. Confirmed appointments are one color and penciled in appointments are another. Confirmation is usually done by sms or call a few days prior to the appointment. Appointments are placed in the calendar with notes and sometimes color tags. Any update is automatically visible to all branches.

9. How do you handle appointment cancellations and rescheduling? What's that process like?

Assistant: Cancellations and rescheduling of patients are treated like any other patient if it’s once or twice, but repeated cancellation or rescheduling will be noted and patients may be taken down the priority list if it gets abusive.

10. How do you handle walk-in patients or dental emergencies when your schedule is already full?

Assistant: The patient will have to wait for the current patient to be finished but there are cases where it will depend on the dentist, if they can get the current procedure done on time or if the next procedure won't take too long. We can accept walk-in or emergency patients but it will have to depend if the schedule lets them squeeze in. Walk-ins are only accepted when there are no scheduled patients for that hour. We prioritize appointments.

11. Do you send appointment reminders to patients? If yes, how?

Assistant: Yes. patient reminders are sent a few days before via sms or call if a patient requests it. We also message them 24 hours before their appointment.

12. What challenges do you face when responding to patient inquiries or appointment requests?

Assistant: Consistency, since all assistants respond during their working hours and also in their free time, sometimes there are inconsistencies when it comes to the conversation between patient and assistant because responses may vary depending on how each staff handles the inquiry. Other problems include wrong or misspelled names being inputted, some assistants accidentally saving appointments on their personal calendar instead of the shared one, difficulty finding patients due to manual typing, no patient profile system, and staff who are not very familiar with technology struggling with calendar management.

13. What tools, apps, or systems do you currently use to manage appointments, records, and billing?

Assistant: google calendar for appointment keeping. Viber/Messenger and sms for communication. google sheets for payment records. Patient records are still written on paper.

14. How are patient records currently stored? Walk me through where information lives and how it's organized.

Assistant: patient records are kept in a mini-booklet format. Patient forms are kept alphabetized and inside filing cabinets in a storeroom. Each record includes contact details, medical history, procedures done, and their consent forms.

15. What information is typically included in a patient's record?

Assistant: contact information, past and present medical and dental history, procedure done on the patient inside the clinic.

16. When you need to find a patient's record, what's that process like? Who does it, and how long does it take?

16.1 Probe: If a patient returns for a follow-up visit, how do you check what was done during their previous visits?

Assistant: Assistants do it. They'll need a few minutes to find it since it’s alphabetized. Patients with follow up checkups are usually not put back in the cabinet especially if the follow up is within the month. There's a rack for upcoming patients and recent patients. It takes around 5-10 minutes depending on how organized the filing is. Assistants retrieve records before the patient's procedure and update them after payment and treatment. Patients are also required to fill out parts of the booklet themselves upon arrival.

17. Have you experienced situations where records were hard to find, incomplete, or had conflicting information? Can you give an example?

Assistant: Yes. Since it’s paper, sometimes records are misplaced in a wrong drawer or somewhere else. Also, the limited space in the booklets restricts how much information the dentist can record about findings, procedures, and treatments.

18. After a patient finishes their treatment, what happens next with billing and payment? Do all patients pay in full right away?

18.1 Probe: How do you keep track of who still owes money and follow up with them?

Assistant: The dentist will charge the patient, and assistants will handle the billing. Payments depend on the procedure. Some procedures can be partially paid and some have to be paid in full. Partial payments are noted through paper notes attached in the patient form and patients are reminded of their next appointment and balance. After the dentist finishes the procedure, the patient is shown a breakdown of what was done and the fees, then pays and is issued an invoice. Patients are informed of the estimated price before treatment, but the dentist finalizes the exact cost after the procedure.

19. When you're checking that all payments for the day were recorded, what's that process like? How do you catch errors or missing entries?

Assistant: everything is inputted in google sheets, expenses and sales. The money is then double checked if they missed anything or if everything’s in place.

There is no digital system that automatically tracks unpaid amounts, so staff must remember them manually. When a patient pays part of their balance, the record has to be located again to update the balance.

20. What problems do you run into with billing or payment tracking?

Assistant: Sometimes there are errors in noting what payment method the patient used thus creating a problem balancing the numbers, and sometimes assistants forget to remind the patient regarding their partial payments. For procedures like braces, patients do not pay in full and if they do not return for months, it becomes hard to track because unpaid balances are not automatically reflected in any system. There was also a time that a patient paid with a check and the check did not go through, but since it was an old patient the check was reissued. Verifying cashless transactions can also take extra time if the proof of payment is misplaced or unclear.

21. What are the three biggest problems related to managing appointments, patient information, or payments that you deal with repeatedly?

21.1. Have different staff members encountered similar issues as yours?

Assistant: double booking for appointment issues. patient information doesn’t match their HMO for patient information. none so far on payments but assistants sometimes forget inputting expenses.

22. If we built a system for your clinic, what features or capabilities would be most valuable to you?

Assistant: putting everything in one app or system because currently we’re using multiple platforms. Another is maybe creating it for mobile use because some of the platforms we use are hard to use on mobile devices. Easy use for staff that aren't that knowledgeable on technology and easier access to patient records and their payment.

23. Is there anything about managing the clinic day-to-day that we haven't talked about yet that causes frustration or takes up a lot of time?

Assistant: Yes. It’s really tiring sometimes. We have to check messages, patient records, and the calendar all at the same time. Sometimes we have to type the same info in different places like the paper booklets and Google Sheets. Finding records can take a while, and sometimes things get misplaced. Because everything is in different tools, it takes longer to see what’s going on in the clinic. The paper booklets also don’t have enough space, so we can’t write everything down. All of this makes the work slower, and we spend a lot of time just managing the info instead of helping patients and dentists.

SECOND INTERVIEW

Mrs. Michelle A. Macaisa 04/22/26
Date & Signature Over Printed Name
Position: Owner/Dentist

Ms. Andrea Macaisa 04/22/26
Date & Signature Over Printed Name
Position: Dentist

Czarina Kiel Macaisa 04/22/26
Date & Signature Over Printed Name
Position: Assistant

24. Is your clinic operating in multiple branches?

Owner: Yes, we operate in multiple branches. Our locations are at Circle C (the main branch), Antipolo, and the Commonwealth branch.

24.1 If yes, which branch would you prefer for the initial implementation of the system?

Owner: I would like the system to be implemented in the Commonwealth branch first. Once it is working properly, you can proceed with implementing it in the other branches.

24.2 When was the business started?

Owner: The business started way back 2002.

25. On average, how many patient inquiries do you receive per day across all platforms?

Assistant/ Dentist: On normal days, we receive around 10–20 inquiries per day across SMS, Messenger, and Viber. During peak seasons like summer, Christmas, and especially February (Valentine’s season), it increases to around 20–30 inquiries daily, since many patients try to book appointments for aesthetic procedures, but we can only accommodate a limited number of patients per day.

26. Last time you mentioned that assistants handle appointment scheduling, patient records, and billing and payments. With two assistants working at the same time, how do you divide those responsibilities — do you each handle all three, or do you each take on specific tasks?

Assistant/ Dentist: There are no strict rules about which particular assistant will do particular tasks. In a situation where there are two (2) dentists and two (2) assistants, almost all the time, the assistant who could do the task immediately would be the one to do it, considering all assistants are trained to handle all front-desk tasks and assisting the dentists.

26.1. Does the division change depending on how busy the day is, or is it fixed?

Assistant: Both assistants and dentists can do all the tasks that the front desk can do, there are no specific tasks that the assistants are assigned into. If there are no assistants to handle the front desk duties then the dentist will handle those tasks.

26.2. Has the split ever caused something to be missed because each of you thought the other was handling it?

Assistant: Yes, since scheduling is handled by all the assistant that was free for a time being there was a doubling booking occurrences, there was a time when I was talking to the patient for scheduling them and I did schedule them but when I was asked by my dentist to help her, then someday later the patient came back or called back and change her mind but it wasn't me she talked to but the other assistant and that assistant didn't know that I already booked her for someday and time so the other assistant booked her on someday and time.

26.3. Does the owner want to hire or want to give the administrative duties to someone else?

Owner: No, I am okay to handle the administrative duties since I am the owner of the clinic. I don't want anyone else to handle my business except me.

27. From the moment a patient first messages you to the moment their appointment is fully confirmed, how long does that usually take — and what causes it to take longer sometimes?

27.1. Is there a specific step that consistently slows things down — like waiting for the patient to reply, or needing to check the calendar with someone else?

Assistant: When the patient calls it usually takes roughly 30 min of our time but as I said earlier the communicating with the patients using sms, viber or messenger depends with how the patient replied back, there are some patient that takes half an hour for them to reply back and some would take hours, we don't ask why we just reply to their messages and sometimes the HMO itself when we are asking for the patient's verification identity they usually take days or weeks when they replied back to us.

28. Roughly how much of your day goes into managing appointments — responding to inquiries, updating the calendar, sending reminders? Does it ever get in the way of assisting during procedures?

Assistant: That depends. When the 2 assistants are busy handling the patient with the dentist then when one of the assistants is finished helping, then that assistant will be the one handling the inquiries or the front desk job.

28.1 Has there ever been a case where 1 dentist needed at least two dentist, and all the inquiries weren't handled right away?

Assistant:

Yes, there are times when both of us assistants are busy helping the dentists during procedures, so we can't respond to inquiries right away. Because of that, some messages get delayed, which also delays the creation and confirmation of appointments.

29. Have you ever encountered a situation where a patient followed up through a different channel from where they originally booked — say they messaged on Viber first, then followed up by SMS — and it caused any confusion or miscommunication?

29.1 What happened, and how did you resolve it?

Assistant: Yes. There was a patient that I was talking to on sms then that person suddenly moved to messenger for some reason and that caused confusion for me because why would you suddenly change the communication platform when you are talking to me straight then I would receive a message on our messenger continuing the conversation like nothing happened. I just went on with the patient I am talking to.

30. You mentioned the calendar uses color codes for appointments.Can you walk us through what each color represents and how that system works day to day?

30.1 Has there ever been a situation where a color tag was misread or applied incorrectly and it caused a problem with a patient's appointment?

Assistant/ Dentist: Technically, in the commonwealth branch when we set a schedule on google calendar, we can choose a color per appointment we create. So, we can easily see if that certain color is for (blue) the patients that already had a schedule but they aren't yet agreeing to said schedule, (pink) confirmed patients, (grey) cancelled patients. Yes, since the two assistants handle the scheduling, inquiries etc. there are times when patient cancelled the schedule then sometime later on the patient comes back and change his/hers mind and thinks of rescheduling it instead but talks with different assistant and booked that patient in another time.

31. You mentioned that HMO patients need to be verified before getting a schedule, and that it takes 1 to 2 days. Why does verification take that long — what exactly is happening or being waited on during those 1 to 2 days?

31.1 Has it ever taken longer than expected? What did you tell the patient while they waited?

Assistant: Yes, sometimes it takes weeks just so we can say to the patient that their HMO is verified and can use it. That's on HMO side not ours, as soon as we get the info of the patient's HMO credentials, we will call the HMO right away to let them verify the info of the patient. We do not know what is happening and why it takes too long for verification. We just tell the patient that the verification process would take 1-2 days or maybe weeks and it is not on us but how the HMO does their verification process. Since the verification takes a long time, that's when we have lost patients but it is not our fault it's the HMO verification process's fault why we lose some patients.

32. You mentioned that patient information not matching their HMO details is one of your top problems. Have you encountered a specific situation where that mismatch caused a real problem? Can you walk us through what happened?

32.1 At what point was the mismatch discovered — during booking or on the day itself? Was the patient turned away?

Assistant: Well, it's not just HMO details that I was talking about it's the mismatch of patient details that are the top problem even way back when I wasn't the assistant here, patient mismatching info occurs over and over again, I just said HMO details because that occurs to me often when the patient often give missing information or a wrong information and when the mismatch is occur during the verification we always tell that to the patient and they would have to wait again for couple of days to be verified and ending we lose those patients who are waiting.

33. You mentioned sending reminders via SMS or calling a few days before, and again 24 hours before the appointment. Is that something you do manually for every appointment, or only when a patient specifically requests it?

33.1 Has a reminder ever not been sent and the patient didn't show up as a result? How often does that happen?

Assistant: Yes, we do that manually every appointment. No, we always remind our patient of their appointment time and day.

34. How many filing cabinets or storage sections do you currently use for patient records, and how are they organized across them?

Assistant: There are many. Each letter has its own drawer, and records are arranged alphabetically by surname. For example all patients with surnames starting with “A” are placed in the “A” drawer, “B” in the “B” drawer, and so on. There are many drawers used for storing patient records. Also, we are not allowed to dispose of any records even if they are already 10 years because it is part of our clinic policy.

34.1 Does having more sections make it faster or harder to locate a specific record? What do you think happens to the system as the number of patients keeps growing?

Assistant: It becomes harder because even if records are arranged alphabetically, we still need to go through many files. As the number of patients grows it takes more time to find records and increases the chances of misplacement.

35. Since you are using physical storage, are there instances where records are tampered with lost, damaged, or exposed to water?

Assistant: Yes, since we are putting the records on the rack before the patients arrive there are many instances where records get damaged. For example, if a record gets wet, we can no longer do anything about it because it is already damaged, and we do not have any backup copies.

36. You mentioned it takes around 5 to 10 minutes to find a patient record depending on how organized the filing is.

Assistant: Even though the records are arranged alphabetically, we still need to check them one by one. For example, if the patient's surname is Santos under the letter "S," there are many "Santos" records, so we still have to go through each one. It becomes more difficult and time-consuming if a record is misplaced because we have to search through multiple files or even different letters.

37. On a day with 6 to 8 patients, do you pull all the records at the start of the day, or one by one as each patient arrives? Are there ever moments where multiple booklets are being handled at the same time?

Assistant: We usually pull the records one by one as each patient arrives, however, this method sometimes causes delay if the record is not immediately found. Though sometimes I pull out the records of the scheduled patients for the month all at once whenever I have time to do it. It is just that when all the booklets are all out and several of them would cancel the appointment, it would take time for me to return the records back and alphabetize them which could be avoided if those were not pulled out if the patients did not arrive.

37.1 Has handling multiple booklets at once ever led to a record being mixed up or put back in the wrong place?

Assistant: Yes, there are instances where records get mixed up to the other files especially when several booklets are being handled at the same time. Sometimes a record is accidentally placed in the wrong cabinets or returned to the wrong section of the cabinet

38. Are there times when records are not returned to their proper drawer (for example, misplaced, lost in the clinic, or taken home by patients)?

Assistant: Yes, there are instances where records are not immediately returned to the drawer, but we are still able to find them later. As for patients taking records, that has never happened because it is not allowed. Patients are not allowed to handle the records, and if they need a copy, they can only take a photo. They are not allowed to bring the records home.

39. You mentioned that records sometimes end up misplaced in the wrong drawer. Can you give us a specific example of when that happened while a patient was already at the clinic waiting — what did you do, and did the procedure get delayed?

Assistant: Yes, like. For example, sometimes a record from letter B gets stuck in the files under letter A. So because of this we have no choice but to search for it until we find it. Our racks can become disorganized, especially when patients return within the same month and their records are temporarily kept outside. When we return them to the cabinet, they should be placed alphabetically, but sometimes they get mixed up or inserted between other files, which causes the problem.

39.1 Is it only one assistant who causes these misplacements?

Assistant: No, all assistants have experienced this issue because there are too many records. Misplacement is unavoidable, especially since we are handling thousands of records, possibly around 20,000. Since everything is paper-based, mistakes like this happen.

40. Since you have thousands of records have you experienced running out of space in your filing cabinets or clinic?

Assistant: Yes. As mentioned earlier, we are not allowed to dispose of records, even if the patient is already deceased or the record is over 10 years.. Because of this policy we eventually run out of space.

40.1 What do you do to solve the limited space problem?

Owner: We just added more cabinets. However, the problem remains because as records continue to increase, we keep buying more storage and misplacement still happens more often.

41. Since some patients are family members or relatives, are there cases where they are no longer recorded?

Assistant: No, each patient must have their own record. It depends on the patient if they choose to provide information, but it is their right since it is a legal document. We use these records as a basis in case there are issues involving the dentist. That is why recording information is important, even for family members, for protection purposes.

42. After a procedure, how do you make sure that the booklet is fully and correctly filled out before it goes back into the cabinet?

Assistant: If we notice missing or incorrect information, we usually correct it by referring back to the dentist or the patient's previous visit. In some cases, we ask the patient again during their next visit to complete or confirm the missing information.

42.1 What happens if you notice later that something was missing or filled in incorrectly? Has that ever affected a patient's follow-up visit?

Assistant: Yes, there have been instances where it affected follow-up visits. When information is incomplete or incorrect, it can slow down the review of the patient's history, since we need to double-check and validate the missing details before proceeding with the treatment.

43. After a procedure, who fills in the record — is it something the dentist communicates to you, or do you fill it in based on what you observed during the procedure?

Assistant: After a procedure the record is usually filled in based on the information communicated by the dentist, since they are the main source of the clinical details.

43.1 Has there been a situation where something was recorded incorrectly because of a miscommunication? How was it caught, and what was the impact?

Assistant: Yes, there are cases where miscommunication between the dentist and staff leads to incorrect recording. Usually, this is caught during checking or when reviewing the patient during their next visit.

44. You mentioned that the limited space in the booklets restricts how much can be recorded. We also understand that when a booklet is full, a second one is created for the same patient.

Assistant: When the booklet becomes full, we attach an additional sheet. We usually photocopy the front page and use it as a continuation. It is important to complete all the required information because it is a legal document.

Patients themselves fill out the forms, and there are no shortcuts when it comes to medical information, so we really need to attach extra sheets when needed.

45. When a patient ends up with two booklets, how do you make sure both are retrieved together every time that patient comes in — and has there ever been a case where only one was pulled and the other was missed?

Assistant: When a patient has two booklets, we try to store and label them together under the same patient record to make sure they are always pulled at the same time during visits. However, because the system is manual and physically stored, there are rare instances where only one booklet is retrieved, especially if the other is placed in a different section or misfiled. In such cases, we have to search the cabinet again until both booklets are found before continuing the procedure.

45.1 Has incomplete information across two booklets ever caused a problem during a procedure?

Assistant: No, there is no issue of incomplete information across two booklets because when the space on the booklet is full, the clinic continues the record using photocopied pages and attaches them using a stapler. However there are times when it still takes longer to find specific details, especially when the record becomes thick or has many pages, which can slightly delay the procedure.

46. For patients undergoing long-term treatments like braces, how do you track their progress across multiple visits inside the booklet? If that patient comes back after several months, can you get a quick picture of where their treatment stands?

Assistant: Sometimes yes, but it is not always fast. We still need to read previous entries in the booklet to understand the patient's progress. If the record is long, it takes time to trace all updates, especially if the visits span several months or years.

47. When a brand new patient arrives for the first time, how long does setting up their booklet and collecting all their information usually take before the dentist can see them? Is there anything in that process that regularly slows things down?

Assistant: The process is sometimes slowed down when patients cannot immediately provide complete medical history or personal information. Also, filling out forms manually takes time especially when there are many patients at the same time.

48: For patients undergoing long-term treatments like braces, how do you track their outstanding balances across multiple visits inside the booklet?

Assistant: It's all written in the booklet. For example, if a patient's braces cost ₱50,000 and they made a down payment of ₱10,000, they have a remaining balance of ₱40,000. After the dentist adjusts the braces, we immediately schedule the next visit — we ask the patient when they're available, say a Saturday next month. When they come back the following month, the dentist adjusts again, and we remind them: "Ma'am, you still have a balance of ₱40,000" because it's written in the record. If they pay, say, ₱2,000, we write it down and update the balance to ₱38,000. The patient then signs to acknowledge that they paid ₱2,000 and that ₱38,000 remains.

49. Is there a required percentage that patients need to pay as a down payment?

Assistant: Yes, the down payment is 50% of the total quotation. The dentist provides the quote based on the patient's case, and we require 50% upfront. The remaining 50% is paid by the patient every adjustment visit.

50. So after the initial 50%, the remaining 50% is divided across the remaining months?

Assistant: Yes. The dentist gives the patient an estimated treatment duration — say two years — so the remaining 50% is divided across 24 months.

51. How do you make receipts?

Assistant: We write them in a receipt booklet — the patient's name, the procedure, the amount, and any applicable discounts like a senior discount, then the total amount to be paid.

51.1 So receipts are also pen and paper?

Assistant: Yes, pen and paper, and we need a carbon copy because we submit the booklets to the BIR.

52. Do you encounter any problems when making receipts?

Assistant: Sometimes there are mistakes, like writing the wrong name. In that case, I get a new receipt, but I still keep the voided one because it still needs to be submitted to the BIR — we keep the original copy.

53. Following up on long-term treatments with the 50% down payment requirement — has there ever been a case where a patient asked to pay less than 50%?

Assistant: Yes, but we don't accept a reduced down payment unless it has been discussed. For example, if a patient approaches the dentist and says "Doc, can I just pay ₱10,000?", and the dentist agrees, then that's fine. However, the dentist still needs approval from the head dentist, who is the owner.

53.1 If that's the case, what if the patient doesn't return or just disappears?

Assistant: We normally just remind them that they have an existing balance. If they really do ignore us, it's ultimately up to them. Technically we could take legal action, but we've never actually seen a doctor sue a patient — it really comes down to their conscience. There are cases where patients come back after quite a long time, say 10 months, and we simply remind them of their remaining balance and still accommodate them.

54. Since you use a shared Google Sheet, have there been any discrepancies among the staff regarding its data?

Assistant: There are times when patients pay directly to the head dentist without the assistants knowing. Until the head dentist informs the staff about the transaction, the patient is still marked as unpaid in Google Sheets from the assistants' perspective. This typically happens with close or long-time patients.

55. When do you record a patient’s payment in Google Sheets?

Assistant: We only record the payment in Google Sheets once the installment is fully completed.

56. How do you track which patients have paid in Google Sheets, and how do you know what treatments or procedures were done?

Assistant: The Google Sheets record contains the patient's name and the total amount they paid only.

57. How do discrepancies occur when recording payments for the day?

Assistant: At the end of the day, we retrieve all the patient records for everyone we accommodated and individually check the computations made by the dentists. Sometimes dentists make mistakes when summing up the total amount, so we need to update the payment records accordingly.

58. When do you record a patient’s payment in Google Sheets?

Assistant: We only record the payment in Google Sheets once the installment is fully completed.

APPENDIX F. PERSONAL TECHNICAL VITAE

Curriculum Vitae of

LLOYD THEO T. ABAD

Blk15 Lot 6C Carnation Street, Dahlia Avenue, Quezon City, Metro Manila, NCR, Philippines
abad.lloydtheo54@gmail.com
+639475888006

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023 - 2027	STI College Fairview
Senior High School	2021 - 2023	STI College Fairview
High School	2020 - 2021	West Fairview High School
	2019 - 2020	First Mile Christian School
	2017 - 2019	West Fairview High School
Elementary	2014 - 2017	First Mile Christian School
	2013 - 2014	Peter and Paul Elementary School
	2012 - 2013	Bethel Christian School
	2011 - 2012	Peter and Paul Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
March 2026	Tagisan ng Talino - Codefest competition	STI College Fairview
March 2025	Tagisan ng Talino - Codefest competition	STI College Fairview

AFFILIATIONS

Inclusive Dates	Name of Organization	Position

SKILLS

SKILLS	Level of Competency	Date Acquired

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop

Curriculum Vitae of

JULIAN PAUL S. ALZATE

1592 jasmin street, Barangay 175, area A, Camarin, Caloocan City, Metro Manila, NCR, Philippines
jpalzate07@gmail.com
+639273347700

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023 - 2027	STI College Fairview
Senior High School	2021 - 2023	Lagro Senior High School
High School	2017- 2021	Maligaya High School
Elementary	2011-2017	Camarin D Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization

AFFILIATIONS

Inclusive Dates	Name of Organization	Position

SKILLS

SKILLS	Level of Competency	Date Acquired
Java (Fundamentals, Object-Oriented Programming)	Beginner	September 2024
Python (Fundamentals)	Beginner	December 2024
SQL (Fundamentals)	Beginner	December 2025
HTML (Basic Web Development)	Beginner	February 2026

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop

Curriculum Vitae of

TIMOTHY ARIEL M. BASILIO

Lot 16, Black 3, Jupiter street, Sunnyside Heights Subdivision, Batasan Hills, Quezon City, Metro Manila, NCR, Philippines
timothyariel.basilio@gmail.com
+639156370127 / +639202123929

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023 - 2027	STI College Fairview
Senior High School	2021 - 2023	Southeast Asian College Inc.
High School	2017- 2021	Capitol Hills Christian School Inc.
Elementary	2011-2017	Capitol Hills Christian School Inc.

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization

AFFILIATIONS

Inclusive Dates	Name of Organization	Position

SKILLS

SKILLS	Level of Competency	Date Acquired

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop

Curriculum Vitae of

JOHN FRANCIS M. GAYO

88 Purok 3, Manga street, Barangay Culliat, Quezon City, Metro Manila, NCR, Philippines
gayojohnfrancis1@gmail.com
+639777096342

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023 - 2027	STI College Fairview
Senior High School	2021 - 2023	Elias A. Salvador National High School
High School	2017- 2021	Elias A. Salvador National High School
Elementary	2011-2017	Agdangan Central Elementary

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization

AFFILIATIONS

Inclusive Dates	Name of Organization	Position

SKILLS

SKILLS	Level of Competency	Date Acquired

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop